SI Appendix

Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

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This supplement gives the full methods, results and supporting diagnostics behind the article. Part I (Supplementary Methods) describes the synthetic grid benchmark, the grid allocation-risk study and the observed-incidence North Carolina (NC) application, and Part II (Supplementary Results) reports the results of each in that order. Queen and rook results, planning years and spillover regimes are labeled separately throughout. The application is a simulation-based design evaluation, not observed intervention efficacy. All data supporting the findings of the simulation study were generated by simulation. The simulation data and R code are available from the authors on request. The North Carolina death-certificate data used in the application are restricted and are not publicly available; only 58-cluster summaries are reported.

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Part I
Supplementary Methods

1 Simulation setting

1.1 Spatial structure and outcome model

We simulated a 10×10 regular grid of 100 clusters under two definitions of adjacency. Under queen contiguity, two clusters are neighbors if they share an edge or a corner; under rook contiguity, they must share an edge. Queen contiguity is the primary analysis. Rook contiguity is a sensitivity analysis that also keeps continuity with earlier work, whose grids used rook neighbors. For each definition, the binary adjacency matrix was row-standardized to form the spatial weights matrix $\mathbf{W}$, so that $\mathbf{WZ}$ gives, for each cluster, the proportion of its neighbors that are treated, and $\mathbf{WY}$ gives the average outcome among its neighbors. Row-standardization expresses each cluster's spillover exposure as an average over its own neighborhood rather than a raw sum, which would otherwise depend on how many neighbors the cluster has. Within a scenario, the same neighbor type defines $\mathbf{W}$ in the outcome model, in the Isolation Buffer design, and in the fitted model.

Cluster-level outcomes $\mathbf{Y}$ were generated from a spatial lag model with a spillover term,

$$\mathbf{Y} = (\mathbf{I} - \rho\mathbf{W})^{-1} \left[\tau\mathbf{Z} + \mathbf{S}(\mathbf{Z}) + \beta\mathbf{X} + \boldsymbol{\varepsilon}\right], \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}), \tag{S1}$$

where $\mathbf{Z} \in \{0,1\}^{100}$ is the treatment assignment produced by the design under evaluation, τ is the direct treatment effect of interest, $\mathbf{S}(\mathbf{Z})$ is the spillover term defined in Equation S2, ρ is the spatial autoregressive coefficient governing dependence in the outcome beyond that induced by treatment and spillover, $\mathbf{X}$ is the baseline incidence covariate with coefficient $\beta = 1$, and $\sigma = 1$. The spatial multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$ propagates each cluster's shocks through the network of neighbors. We tested $\rho \in \{0, 0.01, 0.20, 0.50\}$. When $\rho = 0$ the multiplier is the identity and the model is a linear regression with independent errors, $\mathbf{Y} = \tau\mathbf{Z} + \mathbf{S}(\mathbf{Z}) + \beta\mathbf{X} + \boldsymbol{\varepsilon}$, in which spatial structure enters only through the spillover term; larger values of ρ induce progressively stronger spatial clustering in the outcome.

We evaluated two spillover regimes,

$$\mathbf{S}(\mathbf{Z}) = \gamma\,\mathbf{WZ} \quad \text{(both arms)}, \qquad \mathbf{S}(\mathbf{Z}) = \gamma\,\mathbf{WZ} \circ (\mathbf{1} - \mathbf{Z}) \quad \text{(control only)}, \tag{S2}$$

where $\circ$ denotes the elementwise product. Under the control-only regime, only untreated clusters respond to the treatment status of their neighbors, which represents classical contamination: an untreated cluster's outcome rises with the share of its neighbors that are treated. Under the both-arms regime, every cluster responds to its neighbors' treatment regardless of its own assignment, so neighboring treatment adds to a treated cluster's own effect. We tested four spillover magnitudes, $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, and five magnitudes of the direct effect, $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, with $\tau = 1$ as the primary scenario throughout.

1.2 Heterogeneous baseline incidence

The earlier small-grid study [Walther et al., 2026] assumed a spatially uniform baseline outcome risk. Cluster-level health outcomes typically vary across space, and the designs studied here differ in whether and how they use that variation. We generated the baseline incidence covariate $\mathbf{X}$, with values in $(0, 1)$, under three mechanisms. Under the independent (iid) mechanism, each X_i is drawn from a Uniform$(0, 1)$ distribution, with no spatial structure. Under the spatially autocorrelated

mechanism, $\mathbf{X} = \Phi\{(\mathbf{I} - \rho_X \mathbf{W}_Q)^{-1}\mathbf{u}\}$, where $\mathbf{u} \sim N(\mathbf{0}, \mathbf{I})$, Φ is the standard normal distribution function applied elementwise, $\mathbf{W}_Q$ is the queen weights matrix, and $\rho_X \in \{0.20, 0.50\}$. Under the Poisson mechanism, a spatially autocorrelated log-rate surface, $\log \boldsymbol{\lambda} = \log(35/100{,}000) + (\mathbf{I} - \rho_X \mathbf{W}_Q)^{-1}\mathbf{u}$ with $\rho_X \in \{0.20, 0.50\}$, generates cluster death counts $C_i \sim \text{Poisson}(\lambda_i P)$, with an equal population of $P = 100{,}000$ in every cluster, so that a cluster at the base rate expects 35 deaths. The observed rates C_i/P are then rank-normalized to $X_i = (r_i - 0.5)/N$, where r_i is the rank of cluster i's rate among the $N = 100$ clusters and tied rates share their average rank. Ties remain at this population size: the Poisson surfaces used here have between 60 and 76 distinct values among the 100 clusters. The base rate of 35 per 100,000 was chosen to be representative of published sudden unexpected death (SUD) incidence: it lies between the SUD incidence rates that Mirzaei and colleagues estimated for US men (49.3 per 100,000) and women (21.7 per 100,000) aged 20–64, close to their average [Mirzaei et al., 2019]. It is not the observed rate of the county data described in the Application section, which come from a different source, age range, period, and case definition. The incidence filters always use the queen weights $\mathbf{W}_Q$, whichever neighbor type the scenario uses, so the incidence surfaces are identical under rook and queen contiguity. The iid mechanism and the two strengths of each of the other two mechanisms give five incidence configurations.

For each configuration we generated $K = 10$ incidence surfaces, $\mathbf{X}_1, \ldots, \mathbf{X}_{10}$, each from its own seed. The same ten surfaces are used by every scenario in that configuration, across neighbor types, outcome-model parameters, designs, and values of τ; the design draws and the outcome noise are shared more selectively, as described in Section 4. Each design is applied to the surface that generates the outcome: assignments for surface k are drawn from $\mathbf{X}_k$, and the outcome and the fitted model for those assignments use $\mathbf{X}_k$ as well. This assumes that the incidence observed when a trial is planned is the incidence that drives its outcome (the matched-surface assumption). The target of each scenario is performance averaged over incidence surfaces; for example, the mean squared error (MSE) is $\text{MSE} = E_{\mathbf{X}} E_{\mathbf{Z}|\mathbf{X}} E_{\boldsymbol{\varepsilon}}[(\hat{\tau} - \tau)^2]$, with $\mathbf{X}$ drawn from the configuration's generator, $\mathbf{Z}$ from the design given $\mathbf{X}$, and $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I})$. Bias, coverage, and power are defined in the same way.

2 Treatment assignment designs

We implemented eight treatment assignment designs. Six form the comparison set of the detailed analysis and are described here. The other two, Saturation Quadrants and Balanced Halves, are variants of retained designs. Section 2.1 defines them and explains why they were left out of the detailed ranking.

Table S1 describes the six retained designs, and Figure S1 shows one assignment from each, and from SRS, on a single illustrative incidence surface. That surface, spatially autocorrelated with $\rho_X = 0.5$ and generated with a fixed seed, is not one of the simulation's incidence surfaces. Checkerboard is the strict block-stratified allocation of the earlier small-grid study, in which no two rook neighbors share an arm. The earlier small-grid study's Block Stratified Sampling design also admitted relaxed allocations on its 3×3 grid, where the arms cannot be balanced; on its 2×4 and 3×4 grids, as on the 10×10 grid here, the strict allocations are the two checkerboard patterns. We use the two names interchangeably for this allocation and use Checkerboard in the rest of the detailed analysis. Checkerboard is the only deterministic design: it gives the same assignment in every draw, although each of its fits still receives its own outcome noise. Every other design was drawn independently 25 times for each incidence surface. Designs that rank clusters by incidence break ties at random, afresh for each draw. High Incidence Focus is random only through that tie-breaking; under the iid and spatial mechanisms incidence values are distinct, so it gives the

same assignment in all 25 draws on a surface, and its assignment changes only from one surface to the next. Isolation Buffer builds its assignment from the neighbor type of the scenario, and it treats 38 clusters on average under rook contiguity and 21 under queen contiguity. Every other design treats exactly 50.

Table S1: The six treatment assignment designs compared in the detailed analysis, ordered by mean squared error under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (Table S7). Incidence rank is a cluster's rank by the baseline incidence surface $\mathbf{X}_k$, with ties broken at random in every draw. Clusters treated is per assignment; for Isolation Buffer it is the mean under each neighbor type.

Design	Assignment rule	Clusters treated	Deterministic
Incidence-Guided Saturation	The grid is divided into its four 5×5 coordinate quadrants. Quadrants are ranked by mean incidence and given saturations of 80%, 60%, 40%, and 20% from highest to lowest mean; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Quartiles	Clusters are divided by incidence rank into four strata of 25. Twelve clusters are treated at random in each stratum, and one more in each of two randomly chosen strata	50	No
Isolation Buffer	Clusters are treated one at a time, each chosen at random from those that are neither treated nor adjacent to a treated cluster, until none remain (a greedy random maximal independent set); no two treated clusters are neighbors	38 rook, 21 queen (mean)	No
High Incidence Focus	The 50 clusters with the highest incidence rank are treated	50	No
2×2 Blocking	The grid is divided into 25 nonoverlapping 2×2 blocks; two of the four clusters in each block are treated at random	50	No
Checkerboard	Clusters whose grid coordinates (x, y) have an odd sum $x + y$ are treated, so treatment alternates from cluster to cluster; under rook contiguity every neighbor of a treated cluster is a control, and every neighbor of a control is treated	50	Yes

2.1 The two related variants

Table S2 defines Saturation Quadrants and Balanced Halves in the format of Table S1. Each shares the mechanism of a retained design, and in the comparison of the eight designs and simple random sampling (SRS) at $\tau = 1$, neither could be distinguished from its counterpart under either neighbor type by the Nemenyi and Holm-adjusted Wilcoxon signed-rank tests described in Section 4.1. We kept one design from each pair for the detailed ranking. Both variants remain in the main-text

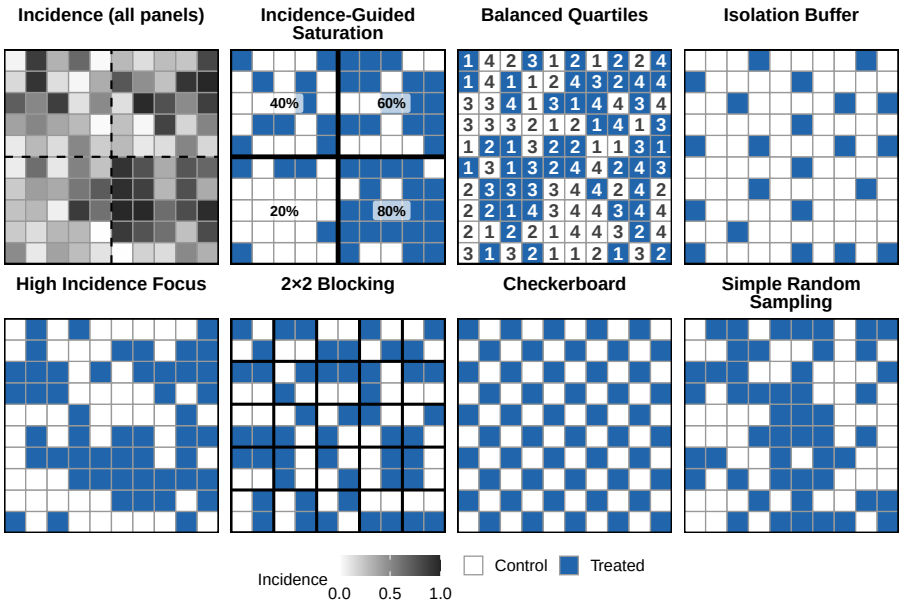


Figure S1: One example assignment from each of the six designs and SRS on the 10×10 grid, in the order of Table S1, all on the illustrative incidence surface in the first panel (darker is higher). Blue clusters are treated and white clusters are controls. Lines mark the 2×2 blocks and the quadrants, numbers in the Balanced Quartiles panel give incidence quartiles, and percentages in the Incidence-Guided Saturation panel give treated shares. Isolation Buffer is shown under queen contiguity.

nine-design benchmark and in the application, and Section 7.4 reports the comparison of all eight designs at the primary scenario, $\tau = 1$.

Saturation Quadrants gives the quadrants their saturations in random order, whereas Incidence-Guided Saturation gives the higher saturations to the quadrants with higher mean incidence. Saturation Quadrants' mean squared error was 0.090, against 0.091 for Incidence-Guided Saturation, under queen contiguity (Nemenyi $p = 0.99$; Holm-adjusted Wilcoxon signed-rank $p = 0.62$), and 0.071 against 0.072 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 1.00$). It also had the better average rank (2.394 against 2.700 under queen contiguity and 2.587 against 2.650 under rook contiguity), so keeping Incidence-Guided Saturation does not rest on performance. Incidence guidance matches the accuracy of a random quadrant order while directing the highest saturations to the quadrants or regions with the highest incidence. How much treatment it redirects depends on how much incidence differs between regions (Figure S20). The accuracy advantage of both designs over SRS, which arises under control-only spillover (Section 12.2), comes from the saturation structure they share. On the NC map, both designs use four fixed geographic regions in place of the quadrants.

Balanced Halves divides clusters into two halves by incidence rank and treats 25 at random in each. Its mean squared error was 0.130, against 0.131 for Balanced Quartiles, under queen contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 1.00$), and 0.084 against 0.084 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 1.00$). The two designs differ only in how finely they stratify on incidence, and we kept the finer, four-stratum version. Balanced Halves had the marginally lower mean and the better average rank under queen contiguity (3.831 against 3.906), and Balanced Quartiles had both under rook contiguity (3.369 against 3.569), so this choice does not rest on

performance either.

Table S2: The two related variants outside the detailed six-design grid ranking, in the format of Table S1. Saturation Quadrants shares its mechanism with Incidence-Guided Saturation, and Balanced Halves with Balanced Quartiles. Incidence rank and clusters treated are defined as in Table S1.

Design	Assignment rule	Clusters treated	Deterministic
Saturation Quadrants	The grid is divided into its four 5×5 coordinate quadrants, which are given saturations of 20%, 40%, 60%, and 80% in an order drawn at random for each assignment, without reference to incidence; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Halves	Clusters are divided by incidence rank into two strata of 50; 25 clusters are treated at random in each stratum	50	No

3 Estimation and identification

We estimated τ with two maximum likelihood spatial lag models, both fit to every simulated outcome and both including an intercept. The primary, oracle model has the spatial lag $\rho\mathbf{WY}$ and the covariates $\mathbf{Z}$, Spill $= \mathbf{S}(\mathbf{Z})$, and $\mathbf{X}_k$, so it matches the outcome model in Equation S1, including the true regime-specific spillover covariate from Equation S2. The oracle model assumes that the analyst knows the form of the spillover mechanism and has the exact exposure term, so its results are a best case for each design; we return to this assumption as a limitation. The non-oracle model, a sensitivity analysis, has the covariates $\mathbf{Z}$ and $\mathbf{X}_k$ only and omits the spillover term. Both models are fit to the same outcomes.

We fit both models with our own implementation of the maximum likelihood estimator of the spatial lag model [Bivand and Piras, 2015], the same estimator that the `lagsarlm` function of the R package `spatialreg` computes with `method = "eigen"`. It maximizes the concentrated log-likelihood over ρ, computing the log-determinant from the eigenvalues of $\mathbf{W}$ [Ord, 1975], estimates the remaining coefficients by least squares given $\hat{\rho}$, and computes standard errors from the analytic information matrix. On 5,120 validation fits it agreed with `lagsarlm` to within 9.7×10^{-8} in $\hat{\tau}$ and to a relative difference of 1.1×10^{-8} in its standard error, at roughly 1% of the computing time; we also refit a 1% sample of the full simulation with `lagsarlm` itself as a check. Confidence intervals are 95% intervals with a t reference distribution and a degrees-of-freedom-corrected standard error, $\hat{\tau} \pm t_{0.975,\,N-p}\,\widehat{\mathrm{SE}}\sqrt{N/(N-p)}$, where $\widehat{\mathrm{SE}}$ is the estimated standard error, $N = 100$ is the number of clusters and p is the number of regression coefficients plus ρ (5 for the oracle model; 4 for the non-oracle model and for rook Checkerboard, whose aliased spillover column is dropped). The application uses the same interval. The 95% Wald interval, $\hat{\tau} \pm 1.96\,\widehat{\mathrm{SE}}$, covered about one percentage point less for every design.

For Checkerboard, strict interspersion makes treatment and spillover hard or impossible to separate. Under rook contiguity on the rectangular grid, $\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$, so treatment and spillover cannot be separately identified; the fitted coefficient on treatment estimates $\tau - \gamma$. Under queen

contiguity, treatment is identified but control-only spillover is weakly separated from treatment. We retain Checkerboard in the main design comparison and flag aliased estimates. Exact matrix identities, boundary information and the rare Isolation Buffer aliases are detailed in Section 3.1. These diagnostics concern the models implemented here.

3.1 Identification diagnostics

Two combinations of design and neighbor type weaken or remove the information that separates τ from the spillover term. Under rook contiguity, Checkerboard gives $\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$ exactly, because every neighbor of a treated cluster is a control and every neighbor of a control is treated. Both spillover covariates then equal $\gamma(\mathbf{1} - \mathbf{Z})$, a linear combination of the intercept and $\mathbf{Z}$, so τ is not identified: the fitting routine drops the aliased spillover column, and the coefficient on $\mathbf{Z}$ estimates $\tau - \gamma$. The earlier small-grid study noted the same consequence of exact collinearity between treatment and spillover exposure, $\hat{\beta} \approx \beta - \psi$ in its notation. The non-oracle model has no spillover column to drop, but its Checkerboard estimates under rook contiguity have the same defect. Under queen contiguity, each of the 64 interior clusters has four treated and four control neighbors, so $\mathbf{WZ} = 1/2$ at every interior cluster. Under the control-only regime the spillover covariate then equals $(\gamma/2)(\mathbf{1} - \mathbf{Z})$ at those clusters, a linear combination of the intercept and $\mathbf{Z}$, and only the 36 clusters on the grid boundary carry information that separates it from the intercept and $\mathbf{Z}$; τ is identified, but imprecisely. Under the both-arms regime it is the constant $\gamma/2$ at interior clusters, confounded with the intercept but not with $\mathbf{Z}$. Isolation Buffer under rook contiguity can also, rarely, draw exactly the checkerboard pattern, a maximal independent set that equals one color class of the grid. This happened in 4 of its 40,000 rook draws and affected one fit in each of 20 scenarios, 4 at each value of τ. We flag every fit in which the fitting routine reports an aliased column and every scenario in which some draw has $\text{rank}([\mathbf{1}, \mathbf{Z}, \mathbf{WZ}]) < 3$. Aliased estimates are kept in all summaries. In the exhibits, we flag Checkerboard under rook contiguity, where every fit is aliased; the Isolation Buffer scenarios, each with one aliased fit among its 250, are noted here and carry no flag in the exhibits.

4 Simulation design, resampling and metrics

The full simulation crossed five values of τ, five incidence configurations, two neighbor types, four values of ρ, four values of γ, two spillover regimes, and eight designs, for $5{\times}5{\times}2{\times}4{\times}4{\times}2{\times}8 = 12{,}800$ scenarios, of which the six designs reported in the detailed analysis account for 9,600. We call a combination of incidence configuration, neighbor type, ρ, γ, and spillover regime a block; there are $5 \times 2 \times 4 \times 4 \times 2 = 320$ blocks, 160 for each neighbor type. At the primary scenario, $\tau = 1$, the six designs give 1,920 scenarios, and every block contains all six.

Within each scenario, we drew $J = 25$ assignments from each of the $K = 10$ incidence surfaces, for 250 fits per scenario, and gave every fit its own noise vector, so that fit (k, j) has outcome $\mathbf{Y}_{kj} = (\mathbf{I} - \rho\mathbf{W})^{-1}[\tau\mathbf{Z}_{kj} + \mathbf{S}(\mathbf{Z}_{kj}) + \beta\mathbf{X}_k + \boldsymbol{\varepsilon}_{kj}]$. Checkerboard's 250 fits share one assignment but have 250 distinct noise vectors. Given a surface, its 25 pairs of assignment and noise are independent, and the ten surfaces are independent of one another, so the surface is the primary sampling unit. Each estimator was fit 3.2 million times. SRS assigns exactly 50 of the 100 clusters to treatment by complete randomization, ignoring incidence and geography. It was run afterward with the same matched surfaces, noise keys and 250-fit structure, bringing the complete benchmark to 14,400 scenarios and 3.6 million fits per estimator, and the original eight-design results were unchanged.

Every random draw was seeded from a hash of a key that names the draw: the configuration and surface index for an incidence surface; the block, surface, and design for a set of 25 assignments; and the block for the noise. Each draw therefore depends only on its key, and the results are identical whether scenarios are run sequentially, in parallel, or in any order. The keys also give common random numbers. Within a block, all designs share the same noise vectors and incidence surfaces; the five values of τ reuse the same assignments and noise; and the oracle and non-oracle models are fit to the same outcomes. Assignments and noise are independent across blocks, but blocks within an incidence configuration share its ten surfaces.

For each scenario and estimator, we computed five metrics over the valid fits, those with a finite estimate and standard error: bias, the mean of $\hat{\tau} - \tau$; the standard deviation (SD) of $\hat{\tau}$; mean squared error, MSE $= \text{mean}\{(\hat{\tau} - \tau)^2\}$; coverage, the fraction of fits with $L \leq \tau \leq U$, where L and U are the limits of the 95% interval defined in Section 3; and power, the fraction of fits with $L > 0$, a one-sided test of $\tau = 0$ at level 0.025. For both estimators, every one of the 250 fits in every scenario was valid, and the fitting routine raised no warning other than the aliasing flags described above.

Each metric is estimated from a finite number of simulated datasets, so it carries Monte Carlo error, which its Monte Carlo standard error measures and which shrinks as more datasets are simulated. We computed these standard errors from the surface-level metrics. For a metric m, we computed m_k on the 25 fits from surface k and set $\text{SE}(m) = \text{sd}(m_1, \ldots, m_{10})/\sqrt{10}$, with intervals $m \pm t_{0.975,\,9}\,\text{SE}(m)$. A difference between two designs is paired at the surface level: its standard error is that of $m_{d,k} - m_{d',k}$ over k. For a metric pooled over the five incidence configurations, we apply the same formula to the 50 configuration-by-surface means, each averaged over that surface's 32 blocks; these 50 means are independent of one another. Section 4.3 gives the parameter grid, the seed keys, and the exact formulas for these metrics and standard errors.

4.1 Statistical comparisons and reporting

We compared designs within each neighbor type with a Friedman rank-sum test on scenario MSE [Friedman, 1937], treating the 160 blocks as repeated measures across the six designs and SRS. SRS is the usual default for a cluster randomized trial, so it enters every test as a seventh member and each design is compared with it as well as with the other designs. At $\tau = 1$ we followed the Friedman test with Nemenyi post-hoc comparisons [Nemenyi, 1963, Demšar, 2006] and paired Wilcoxon signed-rank tests [Wilcoxon, 1945] with Holm's adjustment for multiple comparisons [Holm, 1979] for all 21 pairs, and we repeated the Friedman test at each other value of τ. Blocks within an incidence configuration share their surfaces, so these tests treat dependent blocks as independent, which overstates the information about variation from surface to surface. We therefore also ran paired Wilcoxon signed-rank tests over 50 independent units, the 5 configurations times 10 surfaces, with each unit's MSE averaged over that surface's blocks, and t_{49} intervals for the mean difference. These tests compared each pair of designs adjacent in the mean-MSE order, and each design with SRS within each spillover regime, with Holm's adjustment over the eight designs compared with SRS.

We report results separately for each neighbor type, with queen contiguity primary, and no number is averaged over the two. Unless stated otherwise, every number in the Supplementary Results is pooled over the five incidence configurations, and every table and figure caption states whether it pools the configurations or shows them separately. We report the ranking within each configuration separately and show the split wherever a conclusion changes.

4.2 Allocation-risk study

A separate allocation-risk study used queen weights, $\tau = 1$, all ten surfaces from each incidence configuration and the extreme combinations of ρ and γ, $\rho \in \{0, 0.5\}$ and $\gamma \in \{0.5, 0.8\}$. The risk of assignment a is its conditional MSE $m(a) = E_{\boldsymbol{\varepsilon}}[(\hat{\tau} - \tau)^2 \mid a, \mathbf{X}]$. We estimated it by simulating repeated outcomes for each sampled assignment and then compared its distribution across assignments. This distribution differs from that of squared errors in a single allocation and from MSE across parameter scenarios. The study used 100 assignment draws for each randomized design and 100 simulated outcomes per distinct assignment. One setting (surface 1, $\rho = 0.5$, $\gamma = 0.8$) was refined to 400 outcomes on the same assignments as a precision check; a design with only one possible assignment received its own outcome replication.

4.3 Simulation mechanics

This subsection gives the parameter grid, the random-number streams, the Monte Carlo structure, and the exact formulas for the metrics and their Monte Carlo standard errors.

Parameter grid. Table S3 lists the parameters that were crossed to form the 12,800 scenarios and the settings that were held fixed.

Table S3: Parameter grid of the original eight-design simulation. The crossed parameters form $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 = 12{,}800$ scenarios, 9,600 of them for the six designs of the detailed comparison; the settings in the lower panel are the same in every scenario.

Parameter	Values	Levels
Crossed		
Treatment effect τ	0.8, 1.0 (primary), 1.5, 2.0, 3.0	5
Incidence configuration	iid; spatially autocorrelated, $\rho_X \in \{0.20, 0.50\}$; Poisson, $\rho_X \in \{0.20, 0.50\}$	5
Neighbor type	Queen (primary), rook (sensitivity)	2
Outcome spatial dependence ρ	0, 0.01, 0.20, 0.50	4
Spillover magnitude γ	0.5, 0.6, 0.7, 0.8	4
Spillover regime	Both arms, control only (Equation S2)	2
Design	The six designs of Table S1 and the two of Table S2	8
Fixed		
Grid	Regular 10×10 grid, $N = 100$ clusters	
Outcome model (Equation S1)	$\beta = 1$, $\sigma = 1$	
Poisson mechanism	Base rate 35 per 100,000; $P = 100{,}000$ per cluster	
Incidence surfaces per configuration	$K = 10$	
Assignments per surface	$J = 25$ (Checkerboard: one assignment, used in every draw)	
Fits per scenario and estimator	$KJ = 250$, each with its own noise vector	
Estimators	Oracle (primary) and non-oracle (sensitivity), both fit to every simulated outcome	

Random-number streams. Every random draw in the simulation has its own seed, computed by hashing a key that names the draw. The key lists the draw's type (incidence surface, assignments, or noise) and the parameters that identify it, with numeric values written to a fixed number of decimal places so that equal values always give the same key. The random-number generator and its methods for normal and discrete draws are also fixed explicitly, so the streams do not depend

on software defaults. Table S4 lists the three kinds of draw, the parameters in their keys, and the scenarios that therefore share them. The consequences of this scheme, independence from the order of execution and common random numbers, are described in Section 4.

Table S4: Seed keys. Each key names the kind of draw and the parameters listed here; the incidence mechanism is iid, spatial, or Poisson, and the iid configuration, which has no ρ_X, uses 0 in its place. k indexes the incidence surface and d the design. A draw is shared by every scenario whose key parameters match its own.

Draw	Key parameters	Shared by
Incidence surface $\mathbf{X}_k$	Incidence mechanism, ρ_X, k	Every block, design, and value of τ of the configuration
The $J = 25$ assignments of design d on surface k	Incidence mechanism, ρ_X, neighbor type, ρ, γ, spillover regime, k, d	All five values of τ and both estimators
The 250 noise vectors of a block	Incidence mechanism, ρ_X, neighbor type, ρ, γ, spillover regime	All eight designs, all five values of τ, and both estimators

Monte Carlo structure and checkpoints. Within a scenario, fit (k, j), with $k = 1, \ldots, 10$ and $j = 1, \ldots, 25$, uses surface $\mathbf{X}_k$, assignment $\mathbf{Z}_{kj}$, and noise vector $\boldsymbol{\varepsilon}_{kj}$, the (k, j) column of the block's 100×250 noise matrix (Section 4). The unit of computation is a combination of incidence configuration, neighbor type, and ρ: 40 units, each covering 4 values of γ, 2 spillover regimes, 8 designs, 5 values of τ, 250 fits, and 2 estimators. The units ran in parallel, and the run stops if any unit fails. Each unit's results are saved together with a record of the parameter settings, the simulation code, and the software versions that produced them, and saved results are reused only if all of these match the current run, so results from different code or settings are never combined.

Scenario metrics. For one scenario and estimator, let $f = 1, \ldots, n$ index the valid fits, those with a finite estimate and standard error (here every fit was valid, so $n = 250$). Let $L_f, U_f = \hat{\tau}_f \mp t_{0.975,\,N-p}\,\widehat{\mathrm{SE}}_f\sqrt{N/(N-p)}$, with $N = 100$ clusters and p as defined in Section 3 (the Wald limits replace the multiplier by $z_{0.975} = \Phi^{-1}(0.975) \approx 1.96$), and let $\bar{\hat{\tau}} = n^{-1}\sum_f \hat{\tau}_f$. Then

$$\mathrm{Bias} = \frac{1}{n}\sum_{f=1}^{n}(\hat{\tau}_f - \tau), \qquad \mathrm{SD} = \left\{\frac{1}{n-1}\sum_{f=1}^{n}\left(\hat{\tau}_f - \bar{\hat{\tau}}\right)^2\right\}^{1/2}, \qquad \mathrm{MSE} = \frac{1}{n}\sum_{f=1}^{n}(\hat{\tau}_f - \tau)^2,$$

$$\mathrm{Coverage} = \frac{1}{n}\sum_{f=1}^{n}\mathbb{I}\{L_f \le \tau \le U_f\}, \qquad \mathrm{Power} = \frac{1}{n}\sum_{f=1}^{n}\mathbb{I}\{L_f > 0\},$$

so that $\mathrm{MSE} = \mathrm{Bias}^2 + (n-1)\,\mathrm{SD}^2/n$.

Surface-level standard errors. For $m \in \{\mathrm{Bias}, \mathrm{MSE}, \mathrm{Coverage}, \mathrm{Power}\}$, let m_k be the same formula applied to the valid fits from surface k (here all 25). With $K = 10$ and $s(a_1, \ldots, a_K) = \{(K-1)^{-1}\sum_k (a_k - \bar{a})^2\}^{1/2}$ the sample standard deviation,

$$\mathrm{SE}(m) = \frac{s(m_1, \ldots, m_K)}{\sqrt{K}}, \qquad m \pm t_{0.975,\,K-1}\,\mathrm{SE}(m).$$

Because each surface contributes 25 fits, $m = K^{-1}\sum_k m_k$. For two designs d and d' in the same block at the same τ, the difference is paired at the surface level: with $\Delta_k = m_{d,k} - m_{d',k}$,

$$\mathrm{SE}(m_d - m_{d'}) = \frac{s(\Delta_1, \ldots, \Delta_K)}{\sqrt{K}},$$

with the same $t_{0.975,\,K-1}$ interval.

Standard errors pooled over configurations. For one design, neighbor type, and value of τ, let $\mathcal{B}_c$ be the 32 blocks of configuration c and $m_{b,k}$ the surface-k metric in block b. The configuration-by-surface means and the pooled metric are

$$\bar{m}_{c,k} = \frac{1}{32} \sum_{b \in \mathcal{B}_c} m_{b,k}, \qquad \bar{m} = \frac{1}{50} \sum_{c=1}^{5} \sum_{k=1}^{10} \bar{m}_{c,k},$$

where $\bar{m}$ equals the mean of the metric over the 160 scenarios, and

$$\mathrm{SE}(\bar{m}) = \frac{s(\{\bar{m}_{c,k}\})}{\sqrt{50}},$$

with s taken over the 50 values $\bar{m}_{c,k}$. These are the standard errors in Table S7. For two designs d and d' adjacent in the mean-MSE order (Section 8), let $D_{c,k} = \bar{m}_{d,c,k} - \bar{m}_{d',c,k}$ for $m = \mathrm{MSE}$. The interval for the mean difference is $\bar{D} \pm t_{0.975,\,49}\, s(\{D_{c,k}\})/\sqrt{50}$, and the Wilcoxon signed-rank test, with the normal approximation, is applied to the 50 values $D_{c,k}$.

5 Observed-incidence application methods

5.1 Purpose and observed inputs

The application asks which allocation designs estimate a known education-intervention effect accurately on this geography, when some designs use historical SUD incidence to direct treatment. It supports planning an education intervention for healthcare professionals involved in SUD prevention. There are no observed intervention outcomes and no estimated relationship between local SUD incidence and education ability. Any eventual effect on SUD incidence requires a separate clinical evaluation. A real pilot might realize only one allocation per planning year, making both average estimation error and poor-allocation downside relevant.

The assignment units are 58 community college service-area clusters that partition North Carolina's 100 counties under the study mapping (Figure S2). The map is built from county boundaries and the county-to-college assignment alone, not from the SUD data. Northampton is assigned to Halifax Community College, Bertie to Martin Community College, and Hertford alone to Roanoke-Chowan Community College; these are analysis conventions, not exclusive service commitments. Observed county-year deaths and population estimates for ages 18–64 from the Surveillance, Epidemiology, and End Results program were aggregated to the same named clusters for each year, 2018–2021. SUD deaths were identified from death certificates with the case-adjudication algorithm of Gan and colleagues [Gan et al., 2019]. The corrected numerator supplied by Habib includes heart-failure deaths formerly excluded as presumed non-sudden. It totals 23,523 deaths over 25,594,321 person-years. The earlier numerator, which excluded them, reproduces the totals behind Habib's unpublished report [Habib, 2026]. The application uses the corrected numerator. We do not infer undocumented upstream exclusion rules or zip-code allocation from the received county panel.

Annual incidence is $100,000 D_{iy}/P_{iy}$, and pooled incidence is $100,000 \sum_y D_{iy} / \sum_y P_{iy}$, with the latter population sum interpreted as person-years. All 58 clusters are present in each year, and counts and populations sum back to statewide totals. We hold these observed annual surfaces fixed, generate no synthetic incidence surface and report years separately (Figure S3). The county death-certificate source data are restricted and not publicly available; only 58-cluster summaries are reported.

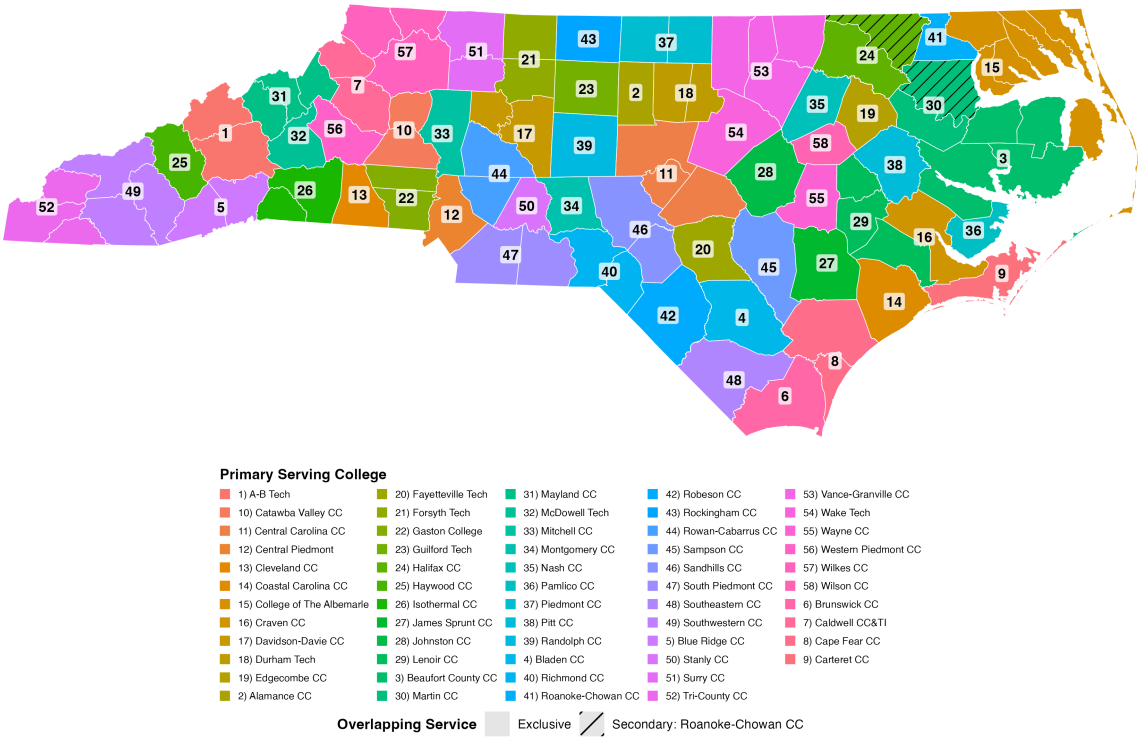


Figure S2: Assignment of North Carolina's 100 counties to the 58 community college service areas used as clusters in the planned application. Each county is colored by its primary serving college, and each service area carries a number keyed to the legend. Hatching marks Northampton County, assigned to Halifax Community College (24), and Bertie County, assigned to Martin Community College (30), which also lie within the broader service network of Roanoke-Chowan Community College (41).

5.2 Geographic adaptations and treatment budgets

Queen contiguity between service areas, computed from legal county boundaries, defines the primary adjacency, and rook contiguity is a sensitivity analysis. Before simulation, we checked the 58-cluster row-standardized contiguity matrix $\mathbf{W}$, its binary adjacency and its row order against the boundary geometry. Legal boundaries shared across water can create adjacency, so $\mathbf{W}$ is a prespecified proxy for geographic exposure, not a measured network of communication among professionals.

All eight designs and SRS keep their grid names, adapted to the map as follows. SRS treats a random 29 of the 58 clusters. High Incidence Focus treats the 29 clusters with the highest annual rates. Balanced Quartiles and Balanced Halves stratify clusters by annual rate rank and treat 29 overall, placing at random the extra treatments that odd-size strata need. In Balanced Halves, one half receives 14 treatments and the other 15, and the half that receives 15 is chosen at random. Isolation Buffer draws a greedy random maximal independent set on the NC adjacency graph. Checkerboard becomes a deterministic greedy graph coloring. Clusters are visited in decreasing order of degree, each receives the arm that conflicts least with its assigned neighbors, equal conflicts are broken toward control, and the assignment is complemented if more than half the clusters are treated. The result keeps Checkerboard's interspersion but is neither a literal rectangular

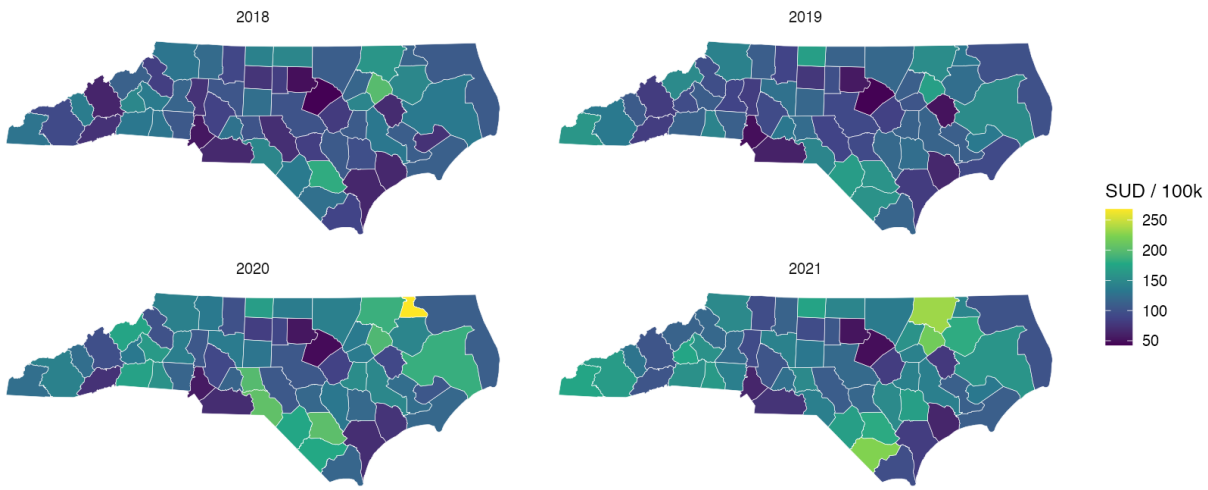


Figure S3: Observed annual SUD incidence per 100,000 adults aged 18–64 in the 58 service-area clusters. These four fixed planning inputs direct incidence-aware allocation; they are not simulated outcomes or measures of education ability.

checkerboard nor an exact bipartite coloring.

The two saturation designs share one fixed four-region partition in all years (Figure S4). A geographic k-means procedure on standardized projected coordinates generated 250 candidate partitions, and we selected one by the balance and compactness criterion $C = 3s_P + 1.5s_K + 1.5s_D + c/d_{\max}$, without consulting outcomes. Here s_P, s_K and s_D are the standard deviations of the regions' shares of reference population, clusters and counties, c averages within-region distances to the regional centers, and $d_{\max}$ is the maximum pairwise distance between the projected cluster coordinates. Reference population is pooled person-years divided by four. Saturation Quadrants gives the four regions saturations of 20%, 40%, 60% and 80% in a random order for each draw. Incidence-Guided Saturation ranks clusters by annual per-capita rate, averages the ranks within each region, and gives 80%, 60%, 40% and 20% to the regions from the highest mean rank to the lowest, breaking regional ties at random for each allocation. Treated counts within each region are rounded under the retained design rules.

Per-capita cluster rates already account for population size, so the regional mean rank weights clusters equally, whereas a pooled regional rate (regional deaths divided by regional population) weights larger clusters more heavily. Mean raw rates and pooled regional rates were prespecified as sensitivity analyses rather than chosen by performance, and Table S6 gives the regional ordering under each summary, documented before performance was evaluated. The grid rule averages $\mathbf{X}$ directly. Mean ranks agree with it in the rank-scaled Poisson configurations but are a deliberate adaptation for the application, not the exact grid rule for continuous incidence.

On the NC map, 2×2 Blocking uses 14 irregular groups of nearby clusters rather than square blocks. The groups are formed once per year by k-means on standardized projected coordinates, incidence ranks and log(1+population), then held fixed, and contain one to eight clusters. Half of each group, rounded, is treated at random. Each year has one single-cluster group, which stays control because half of one cluster rounds to zero, so the yearly totals are 28, 31, 27 and 29.

Treatment budgets (numbers of treated clusters) therefore differ across designs. We report them in Table S5 rather than modify the designs to force equal budgets. Figure S5 shows one example assignment from each strategy.

Table S5: Primary queen application treatment budgets. Entries are mean treated-cluster counts / percent of annual population aged 18–64. Counts and population shares use the same sampled assignments. Population shares are recalculated for each year, including for designs whose assignments do not depend on annual incidence. Shares are geographic reach, not healthcare-professional enrollment.

Design	2018	2019	2020	2021
Checkerboard	27.00 / 50.4	27.00 / 50.4	27.00 / 50.4	27.00 / 50.7
High Incidence Focus	29.00 / 22.2	29.00 / 23.5	29.00 / 23.3	29.00 / 22.4
Saturation Quadrants	29.03 / 49.2	29.03 / 49.2	29.03 / 49.1	29.03 / 49.1
Isolation Buffer	15.89 / 26.7	15.89 / 26.7	15.89 / 26.7	15.89 / 26.7
2×2 Blocking	28.00 / 50.0	31.00 / 55.1	27.00 / 48.0	29.00 / 50.3
Balanced Quartiles	29.00 / 48.5	29.00 / 49.7	29.00 / 50.6	29.00 / 49.8
Balanced Halves	29.00 / 50.0	29.00 / 50.5	29.00 / 49.2	29.00 / 49.4
Incidence-Guided Saturation	29.00 / 53.2	29.00 / 45.1	29.00 / 47.9	29.00 / 51.8
Simple Random Sampling	29.00 / 49.5	29.00 / 49.5	29.00 / 49.5	29.00 / 49.5

Table S6: Regional ordering from highest to lowest summary, using the fixed region IDs in Figure S4. Primary mean ranks weight clusters equally; pooled regional rates weight cluster rates by population.

Year	Mean cluster rank	Mean cluster rate	Pooled regional rate
2018	2,1,3,4	2,4,1,3	4,3,2,1
2019	4,2,3,1	4,2,3,1	4,3,1,2
2020	2,4,1,3	2,4,1,3	4,3,2,1
2021	2,1,4,3	2,4,1,3	4,3,1,2

5.3 Outcome model and Monte Carlo evaluation

For the primary education outcome, we set $\beta = 0$ in Equation S1, use the 58-cluster $\mathbf{W}$, and fix $\tau = 1$, $\sigma = 1$:

$$\mathbf{Y} = (\mathbf{I} - \rho\mathbf{W})^{-1}[\tau\mathbf{Z} + \mathbf{S}(\mathbf{Z}) + \boldsymbol{\varepsilon}], \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}). \tag{S3}$$

Observed incidence affects assignments but neither baseline education nor the constant structural intervention effect. We estimate τ with the same validated maximum likelihood implementation as in the grid study, with an intercept, $\mathbf{Z}$ and the true regime-specific $\mathbf{S}(\mathbf{Z})$, without $\mathbf{X}$. As in the grid model, τ is the coefficient before spatial propagation, not an average total impact after applying $(\mathbf{I} - \rho\mathbf{W})^{-1}$. For the main application and for the analyses of partial participation and sample

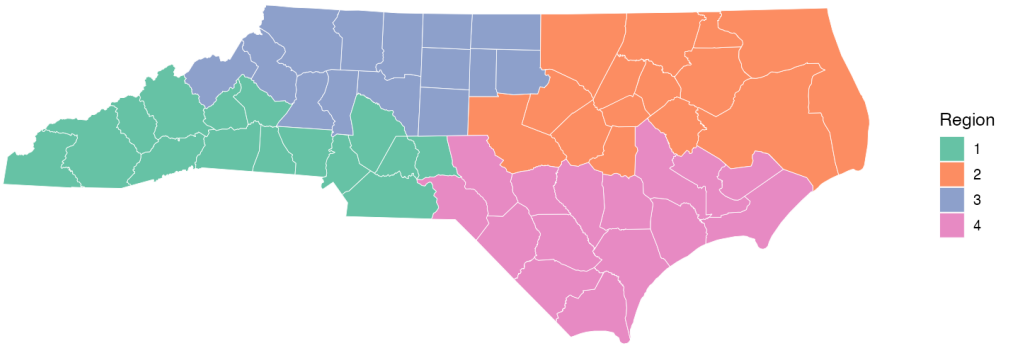


Figure S4: Fixed four-region partition used by both saturation strategies in all years, chosen from 250 geographic candidates before outcome evaluation.

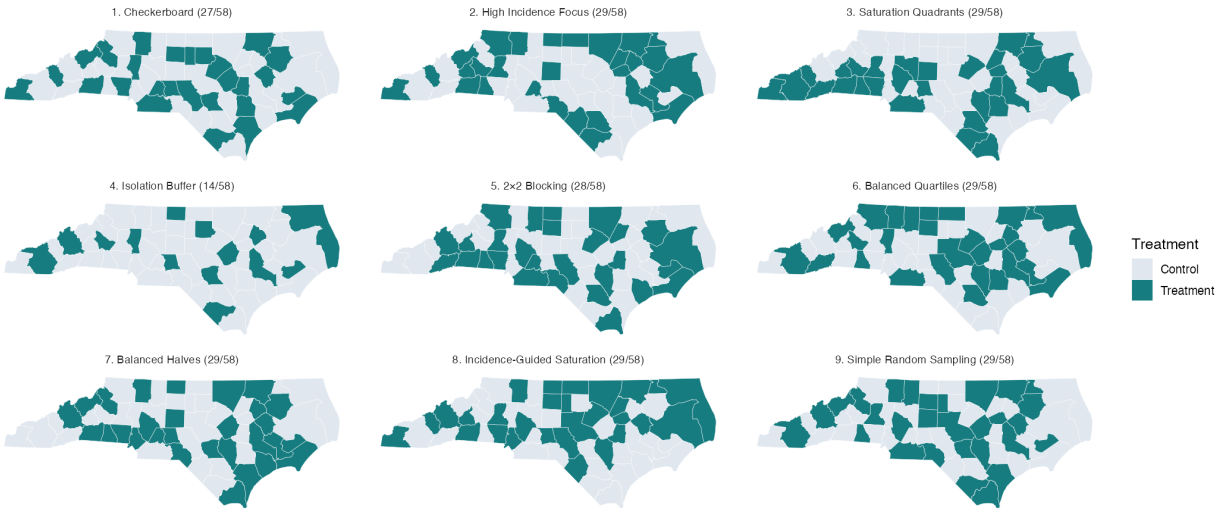


Figure S5: One illustrative queen assignment from each application strategy using 2018 observed incidence and fixed geography. These sampled assignments illustrate the adaptations; they were not selected by observed performance.

size below, we report $\hat{\tau} \pm t_{0.975,\,n-p}\,\widehat{\mathrm{SE}}\sqrt{n/(n-p)}$, where n is the number of participating colleges and p the number of fitted coefficients plus ρ, and we test $\tau = 0$ against the same t distribution. The maximum likelihood variance divides by n, and the normal reference distribution ignores the estimated parameters. With 58 or fewer clusters, both make the Wald interval too narrow.

Primary queen settings cross $\rho = 0, 0.2, 0.5$, $\gamma = 0.5, 0.8$, both regimes, four years and nine designs (432 reported settings). A matched hypothetical baseline sensitivity adds $\mathbf{X} = (\mathrm{rank}_{\mathrm{average}}(rate) - 0.5)/58$ with $\beta = 1$ to both the outcome baseline and the fitted co-

variates (432 settings). It changes baseline and adjustment together and is not an empirical incidence–education association. A rook-contiguity sensitivity analysis of the primary model uses the extreme values $\rho = 0, 0.5$ (288 settings), and the two alternative regional summaries add 96 Incidence-Guided Saturation settings. The full application reports 1,248 settings, computed from 936 distinct simulation settings, because reported settings with the same allocations and outcome model share one simulation.

We initially sampled $J = 100$ assignments and simulated $R = 100$ independent outcomes per distinct assignment. A design whose set of possible assignments contains only one assignment instead received $R = 1,000$ outcomes for that assignment, rather than 100 repeated copies of it counted as separate allocations. An adaptive, prespecified replication schedule extended allocations or outcomes until the Monte Carlo standard error of mean MSE was at most 5% of mean MSE and that of coverage at most 0.01. Two single-assignment settings in the matched sensitivity analysis required $R = 2,000$; all randomized designs met these targets at $J = 100$ and $R = 100$. Each random draw used a fixed, reproducible seed. Settings with the same set of possible assignments shared their simulations, with annual treatment counts and population shares recalculated. In the primary model, Checkerboard, Saturation Quadrants, Isolation Buffer and SRS reuse the same simulations across years, because annual incidence enters neither their allocation nor the outcome model, so their repeated values are not independent annual confirmations.

For each allocation a, the repeated outcomes estimate the conditional MSE $m(a) = E_{\boldsymbol{\varepsilon}}[(\hat{\tau}-\tau)^2 \mid a, \text{observed inputs}]$. We report its distribution across sampled allocations, its estimated 90th percentile and the estimated mean among the worst 10% of sampled allocations, alongside unconditional mean MSE, bias and coverage. An allocation drawn more than once keeps its sampled frequency but uses one conditional estimate, and the Monte Carlo uncertainty accounts for the resulting covariance. A separate tail refinement at queen $\rho = 0, 0.5$, $\gamma = 0.8$, both regimes and all years and designs kept the same allocations and extended the randomized designs to $R = 400$ outcomes per allocation, keeping $R = 1,000$ for single-assignment designs. The refinement is not a new allocation sample. The rank of the fitted design matrix, residual treatment information, aliased columns, warnings, invalid estimates or intervals, and estimates of ρ at the boundary of its parameter space were recorded. Section 5.4 gives the reproducibility and precision details.

5.4 Reproducibility and Monte Carlo precision

The primary application numbers come from the aggregate performance and allocation-risk summaries of the main run and of the separate tail refinement. Annual tables average the six queen parameter pairs equally, without pooled Monte Carlo intervals. Each random draw used a fixed, reproducible seed. An independent check recomputed performance from the stored fits, checked annual treatment budgets and regional orderings, and checked the allocations and outcomes reused from the main run in the tail refinement. The county death-certificate source data are restricted and not publicly available; only 58-cluster summaries are reported.

For randomized designs, the prespecified refinement schedule allowed $J = 100, 200, 400$ and $R = 100, 200, 400, 1000, 2000, 4000$; whichever source of Monte Carlo error, allocations or outcomes, contributed more to the shortfall from the precision targets was extended. All randomized designs met the targets at $J = 100$ and $R = 100$. For designs with a single possible assignment, only outcomes were extended. If an allocation occurs n_a times among J draws, its risk estimate enters the mean with weight n_a/J. The variance due to outcome noise therefore involves squared weights $(n_a/J)^2$, not n_a/J^2 as if duplicate estimates were independent. Bias and coverage use the same accounting of frequencies and covariances. The overall precision includes both variation across assignments and outcome noise given an assignment; a design whose sampled assignments happened

to coincide is distinguished from one with only one possible assignment. Approximate Monte Carlo standard errors of setting-specific differences between designs treat the designs' random draws as independent, whereas averages across settings or models do not assume that shared allocations are independent.

Estimated risk variance is reported raw and with its outcome Monte Carlo noise contribution subtracted. Signed corrected estimates are retained without truncation at zero. None is negative in the NC refinement. Quantiles and worst-decile means use noisy estimated conditional risks. To avoid selection bias, worst-decile assignments were identified with one half of the simulated outcomes and evaluated with the other half, and the roles of the two halves were then reversed. The Jaccard overlap of the two worst deciles describes sampled tail membership, not a confidence interval or population tail probability. For a design with only one possible assignment, an overlap of one is uninformative.

5.5 Partial participation and sample size

All 58 colleges will be invited to take part after the pilot, but not every college will join. We therefore repeated the application for subsets of N participating colleges, $N \in \{15, 20, 25, 29, 30, 35, 40, 45, 50, 58\}$, under the education outcome model with $\beta = 0$. Under random volunteering, 50 subsets were drawn for each N and shared by every design and year, so designs are compared on the same participants. In a sensitivity analysis at $N = 20$ and $N = 29$, a college's chance of joining rose linearly with its incidence rank, so that the highest-burden college was three times as likely to join as the lowest. Each design was rebuilt on the participants with its own rule, and the four saturation regions were required to contain at least two colleges each. Spatial dependence links participating colleges only, so a participant with no participating neighbor has a zero row in the participants' weights matrix $\mathbf{W}_P$. Spillover exposure counts treated participating neighbors over all geographic neighbors, $e_j(\mathbf{Z}) = \sum_{k \in \mathcal{N}(j)} Z_k / |\mathcal{N}(j)|$, so a non-participating neighbor counts as untreated. Spillover scales with the effect, $\gamma = \kappa\tau$ with $\kappa \in \{0.5, 0.8\}$, and the effect τ took nine values from 0 to 2 in units of the college-level outcome standard deviation. Settings crossed these with $\rho \in \{0, 0.5\}$, both spillover regimes, the nine strategies and the four planning years. With fewer than 58 participants, each setting used 20 allocations and 10 outcomes on each of the 50 subsets (10,000 fits), and a design with a single possible allocation on a subset used 200 outcomes instead. With all 58 colleges, each setting used 100 allocations and 100 outcomes, as in the main application, and it reproduced the main application's MSE and coverage within Monte Carlo error. All effect sizes share the same subsets, allocations and noise, so power curves are smooth in τ. For each setting we estimated power with the corrected t test above, the number of colleges needed for 80% power, and the smallest effect detectable with 80% power at each N.

The trial outcome is a score measured on each trained professional and averaged within each college. We write a professional's score as a college effect plus individual variation, with total variance 1, so that effects are in individual standard-deviation units. The intraclass correlation coefficient (ICC) is the share of that variance that lies between colleges, so the between-college variance is the ICC and the within-college variance is $1 - \text{ICC}$. Averaging the scores of m professionals leaves the college part unchanged and divides the individual part by m, so the college mean has standard deviation $\sigma_c = \sqrt{\text{ICC} + (1 - \text{ICC})/m}$ [Donner and Klar, 2000, Hayes and Moulton, 2017]. A college-level effect of τ standard deviations is therefore an individual-level standardized effect $d = \tau\sigma_c$. With $\beta = 0$, multiplying the effect and the noise by the same constant leaves $\hat{\tau}/\widehat{\text{SE}}$ unchanged, so power for any m, ICC and d follows from the simulated college-level results. Rerunning one setting with the effect and the noise both halved reproduced its estimates to within 6×10^{-8}. In a sensitivity analysis, the number of trained professionals grew with college population

(m averaging 25 over participants, ICC = 0.05), so the noise variance differed between colleges while the fitted model stayed unweighted, as an analyst unaware of the difference would fit it.

5.6 Sensitivity analyses

We compared the nine strategies with the Friedman, Nemenyi and Holm-adjusted Wilcoxon tests used on the grid, treating the 24 combinations of year, ρ and γ as blocks, separately for each spillover regime and each value of β. The four years share one map and the strategies whose allocations ignore annual incidence share their simulations across years, so these p-values summarize consistency rather than independent evidence.

To check that the design comparison does not depend on the spatial model, we generated data at $N = 58$ with $\beta = 0$ from the spatial lag model and fitted them by a conditional autoregressive (CAR) model [Besag, 1974, Cressie, 1993] as well as by the spatial lag model. In the reverse direction, we generated data from a CAR model with the same mean and fitted them by both models. We also fitted two Bayesian models to the spatial lag data, a CAR model with the prior of Leroux and colleagues [Leroux et al., 2000, Lee, 2013] and a Bayesian spatial lag model [LeSage and Pace, 2009]. The paragraphs below give the four models, their priors, the Markov chain Monte Carlo settings and the convergence checks. Finally, we replaced the proportional spillover exposure by a binary indicator of any treated neighbor and by a count of treated neighbors scaled by the mean number of neighbors, with the same form in the outcome model and the fit.

Estimation models. Every model in the sensitivity analysis has the mean $\boldsymbol{\mu} = \alpha\mathbf{1} + \tau\mathbf{Z} +$ Spill with the proportional spillover covariate of the main analysis, and uses the 58-college queen geography. The spatial lag model of the main analysis is $\mathbf{Y} = \rho\mathbf{W}\mathbf{Y} + \boldsymbol{\mu} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$. The CAR model is $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{u}$, $\mathbf{u} \sim N\{\mathbf{0}, \sigma^2(\mathbf{I} - \lambda\mathbf{S})^{-1}\}$, where $\mathbf{S}$ is the symmetric matrix similar to the row-standardized $\mathbf{W}$ [Ord, 1975], fitted by maximum likelihood with the `spautolm` function of the R package `spatialreg` [Bivand, 2022]. For the reverse comparison, data were generated as $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{L}\boldsymbol{\varepsilon}$ with $\mathbf{L}\mathbf{L}^\top = (\mathbf{I} - \rho\mathbf{S})^{-1}$ and the same values of ρ as the spatial lag data. These runs used four years, $\rho \in \{0, 0.2, 0.5\}$, $\gamma \in \{0.5, 0.8\}$, both regimes and the nine strategies, with 500 fits per setting.

The Bayesian CAR model [Leroux et al., 2000] is $Y_j = \mu_j + \phi_j + e_j$, $e_j \sim N(0, \nu^2)$, with

$$\phi_j \mid \phi_{-j} \sim N\left(\frac{\lambda \sum_k a_{jk}\phi_k}{\lambda \sum_k a_{jk} + 1 - \lambda},\ \frac{\sigma_\phi^2}{\lambda \sum_k a_{jk} + 1 - \lambda}\right),$$

where a_{jk} is the binary queen adjacency. We used the defaults of the `S.CARleroux` function of the R package `CARBayes` [Lee, 2013], version 6.1.1: independent $N(0, 10^5)$ priors for the regression coefficients, inverse-gamma(1, 0.01) priors for ν^2 and σ_ϕ^2, and a uniform(0, 1) prior for λ. Each chain ran 25,000 iterations, of which the first 5,000 were discarded, thinned by 5. At $\rho = 0.5$, where fewer than half the chains converged, we reran the same data with 100,000 iterations, 20,000 discarded and thinning by 20. The Bayesian spatial lag model [LeSage and Pace, 2009] has the same form as the maximum likelihood model, with the defaults of the `spBreg_lag` function of `spatialreg`, version 1.4.3: a $N(\mathbf{0}, 10^{12}\mathbf{I})$ prior for the regression coefficients, the prior $p(\sigma^2) \propto 1/\sigma^2$, a Beta(1.01, 1.01) prior for ρ rescaled to its feasible interval, and 10,000 draws of which the first 2,000 were discarded. Both Bayesian models used four years, $\rho \in \{0, 0.5\}$, $\gamma = 0.8$, both regimes and the nine strategies, with 100 fits per setting (14,400 fits per model), and both were fitted to the same data as the maximum likelihood model. We report the posterior mean of τ and the 95% equal-tailed credible interval. A chain was counted as converged when the effective sample size of τ was at least 400 and the absolute Geweke z statistic [Geweke, 1992] was below 1.96.

Spillover form. Under the binary form, $\mathbf{S}(\mathbf{Z}) = \gamma\,\mathbb{I}\{\text{any treated queen neighbor}\}$, times $\mathbf{1}-\mathbf{Z}$ under control-only spillover. Under the count form, $\mathbf{S}(\mathbf{Z}) = (\gamma/\bar{d})\times$ (number of treated queen neighbors), where $\bar{d} = 4.93$ is the mean number of queen neighbors, so a college of average degree with every neighbor treated receives γ, as under the proportional form.

6 Estimation error and its dependence on τ

This section explains why the bias of Incidence-Guided Saturation and the coverage of Checkerboard under rook contiguity both grow with τ, as reported in Section 9. The stored results keep neither $\hat{\rho}$ nor the standard error of each fit, so we re-ran every block for the six designs, both neighbor types, and all five values of τ, 2.4 million fits of the oracle model, recording $\hat{\rho}$ and the standard error, and the re-run reproduced the stored results in all 9,600 scenarios. Numbers below that involve $\hat{\rho}$, the standard error, or the terms of Equation S4 come from this re-run; the others come from the stored results.

For one fit (k, j), write $\mathbf{Y}$, $\mathbf{Z}$, and $\boldsymbol{\varepsilon}$ for $\mathbf{Y}_{kj}$, $\mathbf{Z}_{kj}$, and $\boldsymbol{\varepsilon}_{kj}$ (Section 4). Let $\mathbf{V} = [\mathbf{1}, \mathbf{Z}, \mathbf{S}(\mathbf{Z}), \mathbf{X}_k]$ be the 100×4 covariate matrix of the oracle model, so that Equation S1 reads $(\mathbf{I}-\rho\mathbf{W})\mathbf{Y} = \mathbf{V}\boldsymbol{\theta}+\boldsymbol{\varepsilon}$ with $\boldsymbol{\theta} = (0, \tau, 1, \beta)^\top$. Let $\mathbf{A} = (\mathbf{I} - \rho\mathbf{W})^{-1}$ and $\mathbf{V}^+ = (\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$, and for a vector $\mathbf{v}$ write $[\mathbf{V}^+\mathbf{v}]_Z$ for the coefficient on $\mathbf{Z}$ when $\mathbf{v}$ is regressed on the columns of $\mathbf{V}$ by least squares. Given $\hat{\rho}$, the estimated coefficients are $\mathbf{V}^+(\mathbf{I}-\hat{\rho}\mathbf{W})\mathbf{Y}$ (Section 3). Because $(\mathbf{I}-\hat{\rho}\mathbf{W})\mathbf{Y} = \mathbf{V}\boldsymbol{\theta}+\boldsymbol{\varepsilon}+(\rho-\hat{\rho})\mathbf{W}\mathbf{Y}$ and $\mathbf{W}\mathbf{Y} = \mathbf{W}\mathbf{A}(\mathbf{V}\boldsymbol{\theta} + \boldsymbol{\varepsilon})$,

$$\hat{\tau} - \tau = [\mathbf{V}^+\boldsymbol{\varepsilon}]_Z + (\rho - \hat{\rho})\, h + \tau(\rho - \hat{\rho})\, g, \tag{S4}$$

where

$$g = [\mathbf{V}^+\mathbf{W}\mathbf{A}\mathbf{Z}]_Z, \qquad h = [\mathbf{V}^+\mathbf{W}\mathbf{A}\{\mathbf{S}(\mathbf{Z}) + \beta\mathbf{X}_k + \boldsymbol{\varepsilon}\}]_Z.$$

The first term has mean zero given $\mathbf{Z}$ and $\mathbf{X}_k$. The last term has a factor of τ. It is present exactly when $g \neq 0$, where g is the coefficient on $\mathbf{Z}$ when the lagged treatment $\mathbf{W}\mathbf{A}\mathbf{Z} = \mathbf{W}\mathbf{Z} + \rho\mathbf{W}^2\mathbf{Z} + \rho^2\mathbf{W}^3\mathbf{Z} + \cdots$ is regressed on the model's own covariates, and its sign and size are set by the error in $\hat{\rho}$. The concentrated likelihood depends on $\mathbf{Y}$ only through the residual sum of squares of $(\mathbf{I} - \tilde{\rho}\mathbf{W})\mathbf{Y}$ on $\mathbf{V}$, and the part of that residual that involves τ is $\tau(\rho - \tilde{\rho})$ times the residual of $\mathbf{W}\mathbf{A}\mathbf{Z}$ on $\mathbf{V}$. If $\mathbf{W}\mathbf{A}\mathbf{Z}$ lies in the column space of $\mathbf{V}$, that part vanishes for every $\tilde{\rho}$, so $\hat{\rho}$ is the same at every value of τ; if in addition $g = 0$, then $\hat{\tau} - \tau$ is the same at every value of τ. Equation S4 held on every re-run fit, with a largest residual of 4.2×10^{-13}; for Checkerboard under rook contiguity it held in the reduced form given below.

For designs whose treated clusters have treated neighbors, the last term gives a small positive bias that grows with τ. Under queen contiguity, pooled over the 160 blocks, the bias of Incidence-Guided Saturation rose from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$ (Section 9), while that of Isolation Buffer stayed between 0.002 and 0.004. Almost all of the rise is the last term of Equation S4. In our fits, $\hat{\rho}$ was biased downward by about 0.06 on average (for Incidence-Guided Saturation, by 0.071 at $\tau = 0.8$ and 0.054 at $\tau = 3$). We observed this downward bias in these simulations but did not derive it or check it against a published finite-sample result. With $\rho - \hat{\rho}$ positive on average, the last term is positive and grows with τ for any design with $g > 0$. For Incidence-Guided Saturation, g averaged 0.371 over the queen fits, and the last term contributed 0.021 of the bias at $\tau = 0.8$ and 0.043 at $\tau = 3$; the first term contributed 0.001 at both values, and the second 0.007 and 0.002. For Isolation Buffer, g averaged 0.020, and the last term contributed 0.001 and 0.004.

Most of the difference in g comes from the control-only regime. Under the both-arms regime with $\rho = 0$, the lagged treatment is $\mathbf{W}\mathbf{Z} = \mathbf{S}(\mathbf{Z})/\gamma$, one of the covariates, so $g = 0$ for every design. Under the control-only regime, $\mathbf{W}\mathbf{Z} = \mathbf{S}(\mathbf{Z})/\gamma+\mathbf{Z}\circ\mathbf{W}\mathbf{Z}$, and at $\rho = 0$, g is close to the average share

of treated neighbors among treated clusters. That share is 0.57 for Incidence-Guided Saturation, whose treated clusters are concentrated in the high-saturation quadrants, and exactly 0 for Isolation Buffer, whose treated clusters have no treated neighbors. For Isolation Buffer, $\mathbf{Z} \circ \mathbf{WZ} = \mathbf{0}$, so the two regimes define the same spillover covariate and the same g, which is 0 at $\rho = 0$ and about 0.06 at $\rho = 0.5$; at $\rho = 0$, its $\hat{\rho}$ and its estimation error were the same at every value of τ, up to the optimizer's tolerance.

The same term explains why the control-only MSE of Incidence-Guided Saturation rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$ while that of Isolation Buffer did not. Under that regime, averaged over the 80 queen blocks, the variance of the last term of Equation S4 rose from 0.008 to 0.061 for Incidence-Guided Saturation, while the variance of the rest of the error changed little (0.119 and 0.115). The variance of the last term grew more slowly than τ^2 because $\hat{\rho}$ became less variable as τ grew. For Isolation Buffer, the variance of the last term stayed below 0.0002. The rise in bias slowed at larger values of τ for a related reason: the part of the likelihood that involves τ sharpens it around the true ρ as τ grows, so the downward bias of $\hat{\rho}$ shrank.

The decomposition also explains why the coverage of Checkerboard under rook contiguity rose with τ. Under rook contiguity, Checkerboard's coverage rose from 0.150 at $\tau = 0.8$ to 0.464 at $\tau = 3$ (Section 9), although its coefficient on $\mathbf{Z}$ estimates $\tau - \gamma$ at every value of τ. For this design $\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$, and because $\mathbf{W}$ is row-standardized, $\mathbf{W}^2\mathbf{Z} = \mathbf{W}(\mathbf{1} - \mathbf{Z}) = \mathbf{Z}$. The powers of $\mathbf{W}$ applied to $\mathbf{Z}$ therefore alternate between $\mathbf{1} - \mathbf{Z}$ and $\mathbf{Z}$, and $\mathbf{WAZ} = \{(\mathbf{1} - \mathbf{Z}) + \rho\mathbf{Z}\}/(1 - \rho^2)$ lies in the span of the intercept and $\mathbf{Z}$. The derivation of Equation S4 applies with the aliased spillover column dropped, $\mathbf{V} = [\mathbf{1}, \mathbf{Z}, \mathbf{X}_k]$ and $\boldsymbol{\theta} = (\gamma, \tau - \gamma, \beta)^\top$, so that $\tau - \gamma$ takes the place of τ, and $g = -1/(1 + \rho)$. Two consequences follow: $\hat{\rho}$ is the same at every value of τ, and

$$\hat{\tau} - \tau = -\gamma + \delta + (\tau - \gamma)\,\frac{\hat{\rho} - \rho}{1 + \rho},$$

where δ collects the first two terms of that decomposition and, like $\hat{\rho}$, does not depend on τ. The error in $\hat{\rho}$ is therefore multiplied by $\tau - \gamma$, which grows with τ. The mean standard deviation of $\hat{\tau}$ rose from 0.202 at $\tau = 0.8$ to 0.327 at $\tau = 3$, and the standard error tracked it, rising on average from 0.199 to 0.322. With the bias held near $-\gamma$, wider intervals covered τ more often. The rise was largest at $\rho = 0$ (coverage 0.157 to 0.562) and smallest at $\rho = 0.5$ (0.142 to 0.284), consistent with the factor $1/(1+\rho)$. The same term raised the MSE from 0.477 to 0.567. About three quarters of that increase was variance, $(n-1)\mathrm{SD}^2/n$ averaged over scenarios (Section 4.3), which rose from 0.041 to 0.109. The other quarter was squared bias, which rose from 0.436 to 0.458 because $\hat{\rho}$ was biased slightly downward, moving the mean bias from -0.650 to -0.667.

Part II
Supplementary Results

7 Grid benchmark under queen contiguity

7.1 Design performance

At the primary scenario, $\tau = 1$, under queen contiguity and pooled over the five incidence configurations, Incidence-Guided Saturation had the lowest mean squared error (MSE = 0.091), followed by Balanced Quartiles (0.131), Isolation Buffer (0.160), High Incidence Focus (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087) (Table S7, Figure S6). Coverage of the 95% intervals was

between 0.946 and 0.950 for all six designs (Figure S7). Power to reject $\tau = 0$ was 0.921 for Incidence-Guided Saturation, 0.814 for Balanced Quartiles, 0.706 for Isolation Buffer, 0.629 for 2×2 Blocking, 0.582 for High Incidence Focus, and 0.538 for Checkerboard.

Each entry of Table S7 averages the 160 scenarios of one design and neighbor type (5 incidence configurations $\times$ 4 values of ρ $\times$ 4 values of γ $\times$ 2 spillover regimes). The Monte Carlo standard errors of MSE in the table, multiplied by 100, are computed from the 50 configuration-by-surface means (Section 4.3), and those of bias, coverage, and power are all below 0.007. For reference, SRS had an average rank over the 160 blocks of 2.600 under queen contiguity and 2.450 under rook contiguity. The table keeps Checkerboard's rook-contiguity estimates and flags them.

Table S7: Performance of the six designs at $\tau = 1$, pooled over the five incidence configurations, under queen and rook contiguity. Each entry averages 160 scenarios of 250 oracle fits each. Rows follow queen-contiguity mean squared error (MSE); SE of MSE is its Monte Carlo standard error, multiplied by 100. Average rank is the within-block MSE rank among the six designs and SRS (1 = lowest); clusters treated is a mean per assignment. Under rook contiguity Checkerboard is not identified ($\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$), and its coefficient on $\mathbf{Z}$ estimates $\tau - \gamma$.

Design	MSE	SE of MSE ($\times$100)	Bias	Coverage	Power	Average rank	Clusters treated
Queen contiguity (primary)							
Incidence-Guided Saturation	0.091	0.082	0.032	0.947	0.921	1.775	50
Balanced Quartiles	0.131	0.115	0.017	0.946	0.814	2.688	50
Isolation Buffer	0.160	0.112	0.003	0.949	0.706	4.194	21.4
High Incidence Focus	0.245	0.466	0.031	0.948	0.582	5.763	50
2×2 Blocking	0.319	0.319	0.010	0.948	0.629	5.244	50
Checkerboard	1.087	1.011	0.001	0.950	0.538	5.737	50
Rook contiguity (sensitivity)							
Incidence-Guided Saturation	0.072	0.056	0.018	0.952	0.956	1.831	50
Balanced Quartiles	0.084	0.064	0.008	0.948	0.907	2.225	50
Isolation Buffer	0.133	0.091	0.003	0.950	0.784	4.206	38.2
High Incidence Focus	0.206	0.301	0.018	0.948	0.617	5.969	50
2×2 Blocking	0.128	0.124	0.004	0.948	0.803	4.331	50
Checkerboard (flagged)	0.480	0.145	−0.652	0.162	0.390	6.987	50

Incidence-Guided Saturation also led block by block. Of the 160 queen blocks at $\tau = 1$, Incidence-Guided Saturation had the lowest MSE in 95 (59%), Balanced Quartiles in 58 (36%), and Isolation Buffer in 7 (4%); no other design had the lowest MSE in any block.

7.2 Design rankings by incidence configuration

Table S8 gives each design's MSE within each of the five incidence configurations. Under queen contiguity, the mean-MSE order within every configuration was the same as the pooled order: Incidence-Guided Saturation, Balanced Quartiles, Isolation Buffer, High Incidence Focus, 2×2 Blocking, and Checkerboard. The configuration changed each design's MSE only slightly. In this order, each design's highest MSE across the five configurations was lower than the next design's lowest. Incidence-Guided Saturation had the lowest MSE in 18 to 21 of the 32 blocks in each configuration. Coverage did not depend on the configuration either: within each configuration, every design's coverage, averaged over its 32 scenarios, was between 0.941 and 0.952. High Incidence

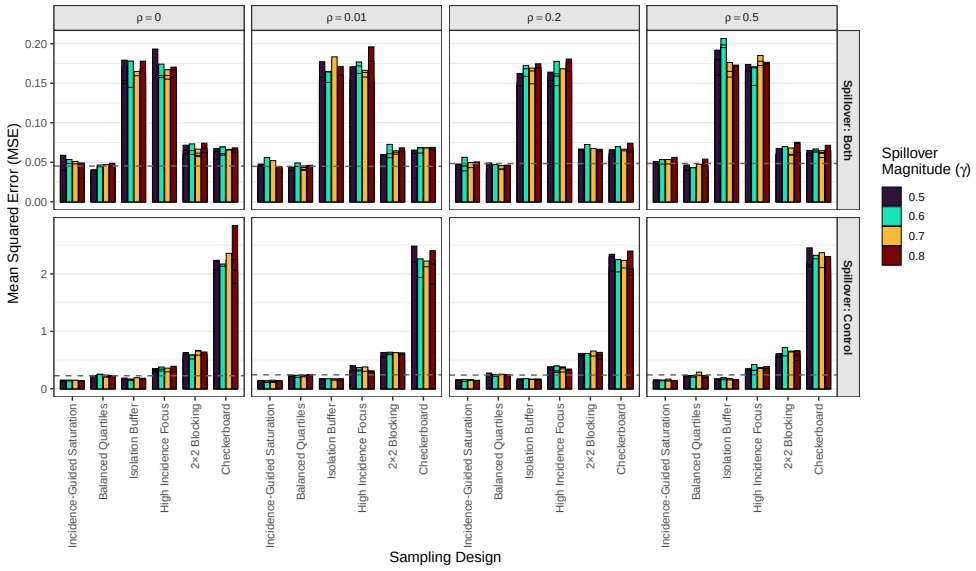


Figure S6: Oracle mean squared error (MSE) by design at $\tau = 1$ under queen contiguity, with designs in the order of Table S7. Columns are the levels of ρ, rows the both-arms (top) and control-only (bottom) regimes, and colors the values of γ. Each bar's height is the largest MSE of the five incidence configurations, and lines within it mark the other four. The gray dashed line marks SRS, the largest of its five configuration MSEs averaged over γ. Vertical scales differ by row.

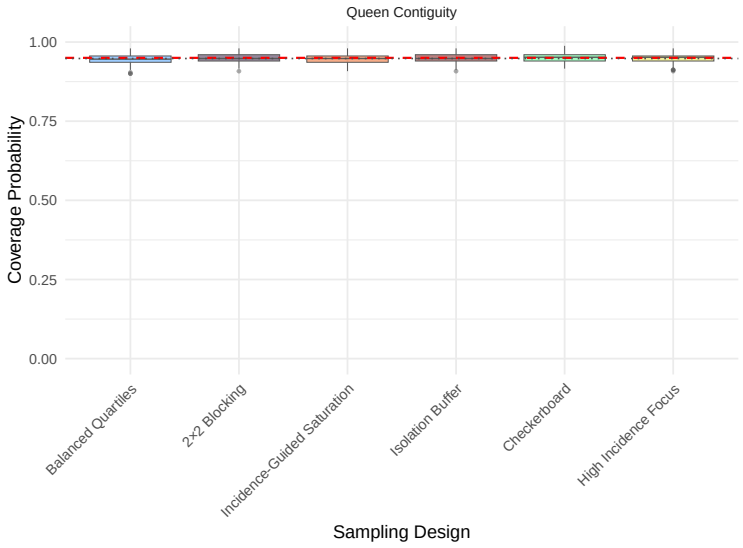


Figure S7: Coverage of the 95% confidence intervals by design at $\tau = 1$ under queen contiguity. Each box summarizes the 160 scenarios of one design, each estimated from 250 fits. Designs are ordered by median coverage, not by MSE. The dashed red line marks the nominal 0.95 level, and the gray dotted line the median coverage of simple random sampling.

Focus, the design that uses the incidence ranks most directly, also had no higher MSE under the two Poisson configurations than under the iid and spatial configurations.

Table S8: Mean squared error by design within each of the five incidence configurations at $\tau = 1$, under queen and rook contiguity. Each entry averages the 32 scenarios of one design, neighbor type, and configuration (4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each from 250 oracle fits. Spatial and Poisson configurations are labeled by ρ_X. Rows follow Table S7. Under rook contiguity, τ is not identified for Checkerboard. Its estimates are kept and flagged.

		Spatial		Poisson	
Design	iid	$\rho_X = 0.20$	$\rho_X = 0.50$	$\rho_X = 0.20$	$\rho_X = 0.50$
Queen contiguity					
Incidence-Guided Saturation	0.093	0.087	0.092	0.090	0.092
Balanced Quartiles	0.130	0.130	0.131	0.128	0.135
Isolation Buffer	0.160	0.159	0.159	0.162	0.161
High Incidence Focus	0.245	0.267	0.245	0.232	0.238
2×2 Blocking	0.311	0.315	0.323	0.326	0.320
Checkerboard	1.091	1.068	1.086	1.112	1.077
Rook contiguity					
Incidence-Guided Saturation	0.072	0.072	0.073	0.073	0.069
Balanced Quartiles	0.081	0.084	0.084	0.084	0.085
Isolation Buffer	0.131	0.131	0.132	0.136	0.134
High Incidence Focus	0.205	0.213	0.200	0.202	0.211
2×2 Blocking	0.129	0.125	0.129	0.130	0.129
Checkerboard (flagged)	0.480	0.482	0.474	0.481	0.482

7.3 Spatial dependence, spillover magnitude and regime

Under queen contiguity, neither the strength of spatial dependence in the outcome nor the spillover magnitude changed the ranking. Across $\rho \in \{0, 0.01, 0.20, 0.50\}$, no design's MSE changed by more than 0.02 (Table S9). Across $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, the MSE of every design except Checkerboard also varied by less than 0.02, and Checkerboard's varied between 1.052 and 1.108 with no trend in γ (Table S10). Under rook contiguity, γ mattered for Checkerboard alone, for the reason given in Section 10.

Table S9: Mean squared error by design across the four levels of outcome spatial dependence ρ, under queen contiguity at $\tau = 1$, pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of $\gamma \times$ 2 spillover regimes). Rows are ordered by pooled mean MSE under queen contiguity (Table S7).

Design	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Incidence-Guided Saturation	0.090	0.087	0.093	0.093
Balanced Quartiles	0.129	0.129	0.132	0.133
Isolation Buffer	0.161	0.157	0.158	0.165
High Incidence Focus	0.243	0.242	0.249	0.248
2×2 Blocking	0.318	0.320	0.314	0.324
Checkerboard	1.095	1.077	1.084	1.092

The spillover regime was the one factor that changed the queen ranking (Table S11, Figure S6). Under the both-arms regime, Balanced Quartiles had the lowest MSE (0.042), followed by Incidence-Guided Saturation (0.046), 2×2 Blocking (0.062), Checkerboard (0.062), Isolation Buffer (0.161), and High Incidence Focus (0.162). Under the control-only regime, the order was Incidence-

Table S10: Mean squared error by design across the four levels of spillover magnitude γ at $\tau = 1$, under queen and rook contiguity, pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of ρ $\times$ 2 spillover regimes). Rows follow Table S7. Under rook contiguity, τ is not identified for Checkerboard. Its estimates are kept and flagged.

Design	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$
Queen contiguity				
Incidence-Guided Saturation	0.092	0.089	0.092	0.089
Balanced Quartiles	0.132	0.128	0.132	0.131
Isolation Buffer	0.159	0.163	0.160	0.159
High Incidence Focus	0.244	0.249	0.248	0.241
2×2 Blocking	0.312	0.314	0.328	0.323
Checkerboard	1.108	1.052	1.086	1.101
Rook contiguity				
Incidence-Guided Saturation	0.069	0.072	0.075	0.072
Balanced Quartiles	0.083	0.085	0.083	0.083
Isolation Buffer	0.131	0.133	0.134	0.133
High Incidence Focus	0.207	0.209	0.205	0.203
2×2 Blocking	0.129	0.129	0.127	0.128
Checkerboard (flagged)	0.295	0.403	0.538	0.684

Guided Saturation (0.135), Isolation Buffer (0.159), Balanced Quartiles (0.219), High Incidence Focus (0.329), 2×2 Blocking (0.576), and Checkerboard (2.111); at larger values of τ, Isolation Buffer matched or passed Incidence-Guided Saturation under this regime (Section 9). For Isolation Buffer, no treated cluster has a treated neighbor, so $\mathbf{WZ} = 0$ at every treated cluster and $\mathbf{WZ} \circ (\mathbf{1} - \mathbf{Z}) = \mathbf{WZ}$. The two regimes in Equation S2 therefore define the same outcome model, and so the same expected MSE; the observed values, 0.161 and 0.159, differ only by Monte Carlo error, because the two regimes use separate draws. Every other design had a higher MSE under the control-only regime, and the increase was largest for Checkerboard and 2×2 Blocking, which went from close to the best designs under the both-arms regime to the two worst under the control-only regime. Among the six detailed designs, Incidence-Guided Saturation was first or second under both regimes, at this and every other value of τ, which matters because a trial team may not know in advance which regime applies. In the full benchmark that includes SRS (Table S16), however, its both-arms MSE was 8% above that of SRS, a small but statistically detectable difference (Figure S15).

The increase under the control-only regime was variance, not bias. The oracle model includes the true regime-specific spillover covariate, so it has no omitted-variable bias. At $\tau = 1$, in every combination of design and regime under queen contiguity, the absolute bias was at most 0.048 and squared bias, averaged over scenarios, was at most 2.2% of MSE. Checkerboard showed the mechanism most clearly: under the control-only regime its spillover covariate is a linear combination of the intercept and $\mathbf{Z}$ at every interior cluster, so only the boundary clusters separate the spillover effect from the treatment effect (Section 3.1). Checkerboard's MSE rose from 0.062 under the both-arms regime to 2.111 under the control-only regime, while its bias stayed near zero (0.004) and its coverage stayed at 0.951, because its intervals widened with its variance. Its pooled MSE of 1.087 was almost entirely variance, and squared bias, averaged over scenarios, was 0.004, less than 0.5% of MSE (Figure S8 shows the decomposition for the Poisson configuration with $\rho_X = 0.20$). 2×2 Blocking showed the same pattern on a smaller scale, with MSE rising from 0.062 to 0.576 and a bias of 0.016 under the control-only regime.

Table S11: Mean squared error by design under each spillover regime at $\tau = 1$, under queen and rook contiguity, pooled over the five incidence configurations. Each entry averages 80 scenarios (5 incidence configurations $\times$ 4 values of ρ $\times$ 4 values of γ). Rows follow Table S7. Under rook contiguity, τ is not identified for Checkerboard. Its estimates are kept and flagged.

	Queen		Rook	
Design	Both arms	Control only	Both arms	Control only
Incidence-Guided Saturation	0.046	0.135	0.045	0.098
Balanced Quartiles	0.042	0.219	0.042	0.125
Isolation Buffer	0.161	0.159	0.131	0.134
High Incidence Focus	0.162	0.329	0.164	0.249
2×2 Blocking	0.062	0.576	0.051	0.206
Checkerboard (flagged under rook)	0.062	2.111	0.477	0.483

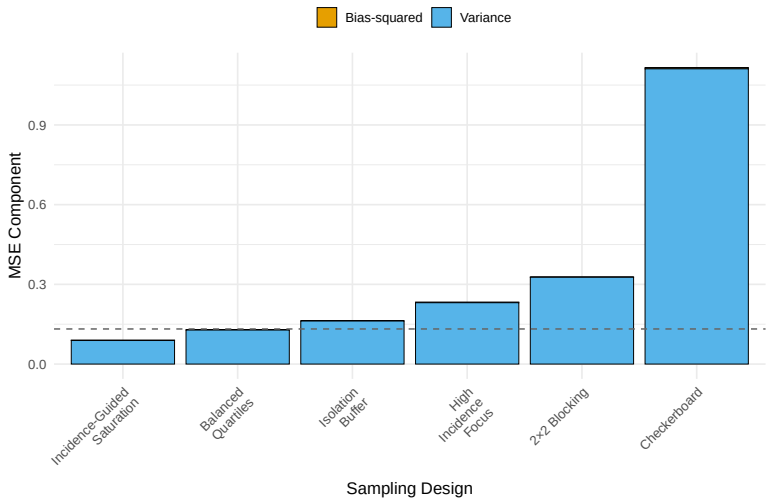


Figure S8: Squared bias and variance (SD^2) of $\hat{\tau}$ by design at $\tau = 1$ under queen contiguity, for the Poisson configuration with $\rho_X = 0.20$. Each stacked component is averaged over the 32 scenarios of that configuration (4 values of ρ $\times$ 4 values of γ $\times$ 2 spillover regimes). Designs follow Table S7, and the gray dashed line marks the MSE of simple random sampling (SRS) in the same configuration.

7.4 The eight-design comparison

Section 2 gives the reasons for leaving Saturation Quadrants and Balanced Halves out of the detailed comparison, with the Nemenyi and Wilcoxon results for each pair. Table S12 reports all eight designs and SRS. Under both neighbor types, each excluded design is adjacent to its retained counterpart in mean MSE. The MSEs and coverages of the six retained designs are those of Table S7, given here to four and three decimal places.

Table S12: Performance of all eight designs and SRS at $\tau = 1$, pooled over the five incidence configurations, under queen and rook contiguity. Each entry averages the 160 scenarios of one design and neighbor type, each from 250 oracle fits. Rows follow queen-contiguity mean squared error (MSE), and Retained marks the six designs of the detailed comparison. Average rank is the mean within-block MSE rank among the eight designs and SRS (1 = lowest). Under rook contiguity, τ is not identified for Checkerboard. Its estimates are kept and flagged.

Design	Retained	MSE	Coverage	Average rank
Queen contiguity (primary)				
Saturation Quadrants	No	0.090	0.946	2.394
Incidence-Guided Saturation	Yes	0.091	0.947	2.700
Simple Random Sampling	Benchmark	0.128	0.948	3.788
Balanced Halves	No	0.130	0.946	3.831
Balanced Quartiles	Yes	0.131	0.946	3.906
Isolation Buffer	Yes	0.160	0.949	5.638
High Incidence Focus	Yes	0.245	0.948	7.763
2×2 Blocking	Yes	0.319	0.948	7.244
Checkerboard	Yes	1.087	0.950	7.737
Rook contiguity (sensitivity)				
Saturation Quadrants	No	0.071	0.949	2.587
Incidence-Guided Saturation	Yes	0.072	0.952	2.650
Simple Random Sampling	Benchmark	0.085	0.947	3.612
Balanced Halves	No	0.084	0.949	3.569
Balanced Quartiles	Yes	0.084	0.948	3.369
Isolation Buffer	Yes	0.133	0.950	6.050
High Incidence Focus	Yes	0.206	0.948	7.969
2×2 Blocking	Yes	0.128	0.948	6.206
Checkerboard (flagged)	Yes	0.480	0.162	8.988

8 Formal statistical comparisons

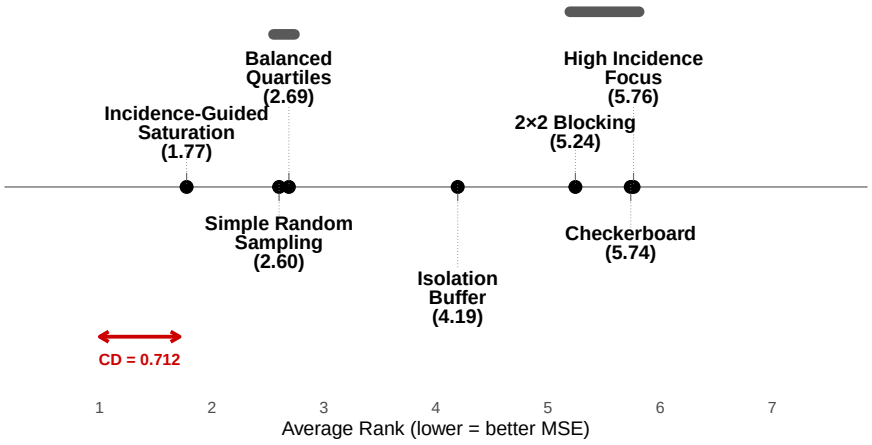
The Friedman test rejected equal MSE ranks across the six designs and SRS under queen contiguity at $\tau = 1$ ($\chi^2 = 560.33$, df = 6, 160 blocks, $p < 2.2 \times 10^{-16}$). The average within-block ranks were 1.78 for Incidence-Guided Saturation, 2.60 for SRS, 2.69 for Balanced Quartiles, 4.19 for Isolation Buffer, 5.24 for 2×2 Blocking, 5.74 for Checkerboard, and 5.76 for High Incidence Focus. Nemenyi comparisons (critical difference 0.712) separated 17 of the 21 pairs (Figure S9 shows the critical-difference diagrams). Incidence-Guided Saturation ranked significantly better than every other design and than SRS ($p = 0.011$). SRS and Balanced Quartiles could not be distinguished from each other ($p = 1.00$), and both ranked significantly better than the remaining four designs. The other three pairs that Nemenyi comparisons did not separate all lay among the bottom three designs: 2×2 Blocking against High Incidence Focus ($p = 0.32$) and Checkerboard ($p = 0.39$), and Checkerboard against High Incidence Focus ($p = 1.00$). Holm-adjusted Wilcoxon signed-rank tests separated 20 of the 21 pairs, all except Balanced Quartiles against SRS ($p = 0.15$).

The bottom three designs were ordered differently by average rank and by mean MSE. By mean MSE the order was High Incidence Focus (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087); by average rank it was 2×2 Blocking (5.24), Checkerboard (5.74), and High Incidence Focus (5.76). Both reversals have the same source: the MSEs of 2×2 Blocking and Checkerboard depended strongly on the spillover regime. Each had a lower MSE than High Incidence Focus in all 80 both-arms blocks (mean MSE 0.062 for each, against 0.162) and a higher MSE in all 80 control-

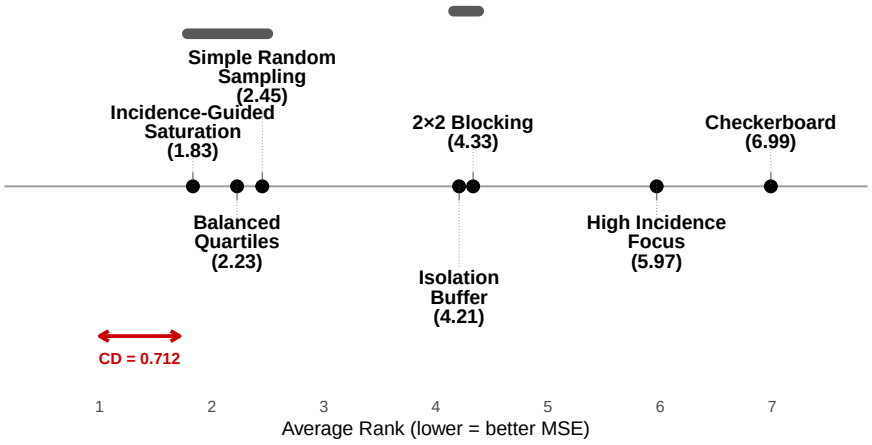
only blocks (0.576 and 2.111, against 0.329; see Section 7.3). Average rank counts each block once, so their wins under the both-arms regime offset their losses under the control-only regime; mean MSE weights those losses by their size.

We also compared each pair of designs adjacent in the mean-MSE order over the 50 independent configuration-by-surface units. Every adjacent pair differed significantly. The mean paired differences in MSE, with t_{49} 95% intervals, were -0.040 (-0.043, -0.037) for Incidence-Guided Saturation against Balanced Quartiles, -0.029 (-0.033, -0.026) for Balanced Quartiles against Isolation Buffer, -0.085 (-0.095, -0.075) for Isolation Buffer against High Incidence Focus, -0.074 (-0.086, -0.062) for High Incidence Focus against 2×2 Blocking, and -0.768 (-0.789, -0.746) for 2×2 Blocking against Checkerboard, with Wilcoxon $p \leq 4.9 \times 10^{-9}$ in every case. The better design of each pair had the lower MSE in all 50 units, except High Incidence Focus against 2×2 Blocking, where it did so in 48 (96%). Because this test follows the mean-MSE order, it separated High Incidence Focus from 2×2 Blocking, which the Nemenyi comparison of within-block ranks did not, because each unit's MSE averages over both spillover regimes, and there the control-only losses of 2×2 Blocking outweigh its both-arms wins.

Figure S9 gives the Nemenyi critical-difference diagrams for the comparison of the six designs and SRS at $\tau = 1$, whose results are reported above for queen contiguity and in Section 10 for rook contiguity. Each design is placed at its average within-block MSE rank among the six designs and SRS over the 160 blocks of its neighbor type. The red arrow shows the critical difference, 0.712 at $\alpha = 0.05$. Under queen contiguity, gray bars join Balanced Quartiles with SRS, and 2×2 Blocking with Checkerboard and High Incidence Focus. Under rook contiguity, they join Incidence-Guided Saturation, Balanced Quartiles and SRS, and Isolation Buffer with 2×2 Blocking. Under rook contiguity, τ is not identified for Checkerboard, and its estimates are kept and flagged.



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure S9: Nemenyi critical-difference diagrams for the six designs and simple random sampling (SRS) at $\tau = 1$, pooled over the five incidence configurations. Designs are placed at their average within-block MSE rank, given in parentheses (1 = lowest MSE); gray bars join designs that the Nemenyi comparisons did not separate.

9 Stability across treatment effect sizes

Under queen contiguity, neither design order changed across the tested values of τ (0.8, 1.0, 1.5, 2.0 and 3.0). The order by average within-block rank was the same at every value, and so was the order by mean MSE, although the two orders differ from each other in the bottom three designs (Section 8; Figure S10a shows mean MSE). With SRS included, the order by average rank placed SRS second at every value of τ, and the Friedman statistic was 564.09, 560.33, 540.92, 526.55, and 497.55 at these five values ($p < 2.2 \times 10^{-16}$ at each). Under rook contiguity the order was also stable (Figure S10b). By average rank among the six designs and SRS, it was the same at every value of τ, with SRS third behind Incidence-Guided Saturation and Balanced Quartiles. By mean MSE, the close pair of 2×2 Blocking and Isolation Buffer traded places, with Isolation Buffer ahead only at $\tau = 3$. The five values of τ share their assignments and noise, so these five comparisons are

not independent confirmations of one another. They show only that, under queen contiguity, the order did not change across the five values of τ we tested, given the same draws. In Figure S10, the shaded bands describe the precision of a single scenario's MSE, not of the plotted average, whose standard error is smaller (Table S7).

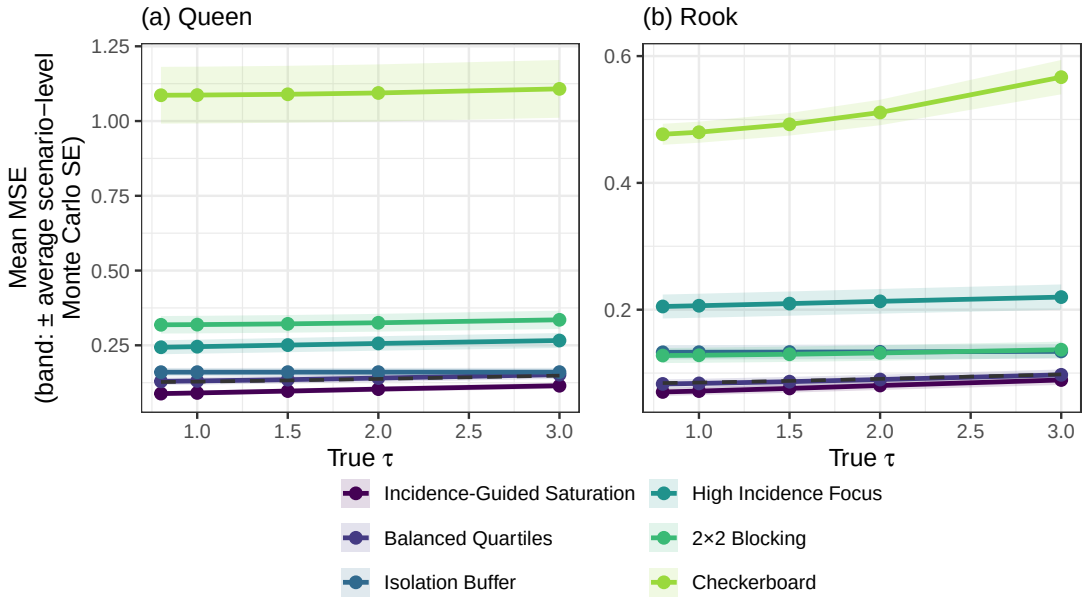


Figure S10: Mean squared error by design across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ under (a) queen and (b) rook contiguity, pooled over the five incidence configurations. Each point averages the 160 scenarios of one design at one value of τ. Shaded bands show ± 1 times the average scenario-level Monte Carlo standard error of MSE. The legend follows queen-contiguity MSE at $\tau = 1$, vertical scales differ, and the gray dashed line marks SRS. Under rook contiguity, τ is not identified for Checkerboard. Its estimates are kept and flagged.

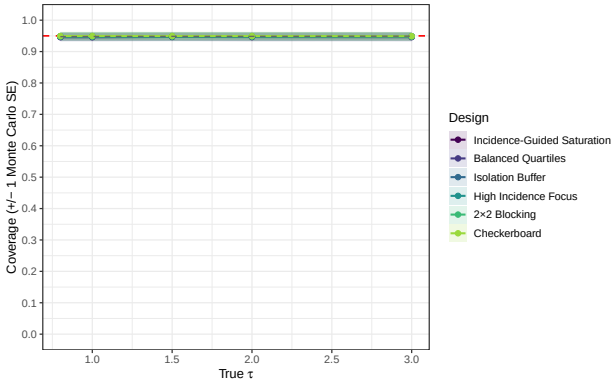
Within that stable order, the margins moved. Under queen contiguity, the average rank of Incidence-Guided Saturation rose from 1.41 at $\tau = 0.8$ to 1.85 at $\tau = 3$, but Balanced Quartiles did not gain on it: Incidence-Guided Saturation had the lower MSE of the two in 103, 102, 102, 102, and 101 of the 160 blocks at $\tau = 0.8$, 1, 1.5, 2, and 3, and the average rank of Balanced Quartiles stayed between 2.13 and 2.14. Isolation Buffer gained instead, and its average rank fell from 3.71 to 3.31. The MSE of Isolation Buffer stayed at 0.160 at every value of τ, while that of Incidence-Guided Saturation rose from 0.088 to 0.115 and that of Balanced Quartiles from 0.130 to 0.152, so the gap between Incidence-Guided Saturation and Balanced Quartiles stayed between 0.037 and 0.041. Isolation Buffer's gain came under the control-only regime, where Incidence-Guided Saturation had a lower MSE than Isolation Buffer in 75, 73, 56, 40, and 12 of the 80 blocks at the five values of τ. The mean control-only MSE of Incidence-Guided Saturation rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$, while that of Isolation Buffer stayed between 0.159 and 0.160 (Section 6). At $\tau = 2$ the two were level (0.160 for Isolation Buffer against 0.160), and at $\tau = 3$ Isolation Buffer had the lower control-only MSE (0.160 against 0.182). Incidence-Guided Saturation remained second under the control-only regime at those two values, second to Balanced Quartiles under the both-arms regime at every value of τ, and lowest in pooled MSE at every value of τ. Its worst case across blocks was the lowest of the six designs at $\tau \leq 1.5$, but at $\tau = 2$ and $\tau = 3$ that of Isolation Buffer was lower (0.207 against 0.210, and 0.208 against 0.238). The mean bias of Incidence-Guided Saturation also

grew with τ, from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$, but squared bias, averaged over scenarios, stayed at or below 3.0% of its MSE, so the rise in its MSE was mostly variance.

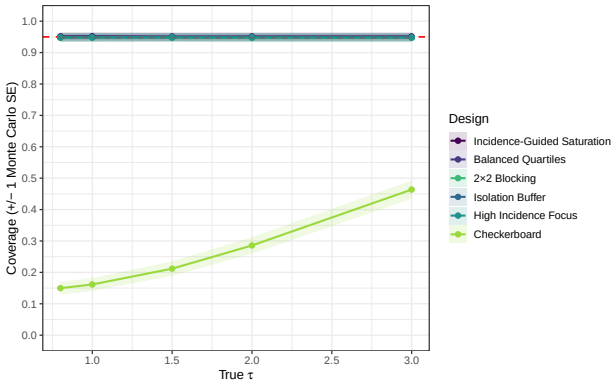
Under rook contiguity, Incidence-Guided Saturation kept the lowest control-only MSE at every value of τ, although its margin over Isolation Buffer narrowed, from 0.095 against 0.134 at $\tau = 0.8$ to 0.131 against 0.136 at $\tau = 3$, where it had the lower MSE in 47 of the 80 control-only blocks. Its worst case across rook blocks was the lowest of the six designs at every value of τ except $\tau = 3$, where that of Isolation Buffer was lower (0.173 against 0.174).

Except for Checkerboard under rook contiguity, coverage stayed close to 0.95 at every value of τ. Pooled over the five incidence configurations, every design's coverage was between 0.946 and 0.950 at every value of τ under queen contiguity, and every design except Checkerboard was between 0.947 and 0.952 at every value of τ under rook contiguity. The coverage of rook Checkerboard, for which τ is not identified, rose from 0.150 at $\tau = 0.8$ to 0.162, 0.212, 0.286, and 0.464 at $\tau = 1$, 1.5, 2, and 3 (Section 6). Figure S11 shows coverage across τ for each design.

In Figure S11, the shaded bands show ± 1 times the average of the scenario-level Monte Carlo standard errors of coverage. They describe the precision of a single scenario's coverage, not of the plotted average. The legend lists the designs in order of pooled mean MSE at $\tau = 1$ for the neighbor type of the panel (Table S7). The vertical axis runs from 0 to 1, so lines close to 0.95 overlap: all six under queen contiguity, between 0.946 and 0.950, and all but Checkerboard under rook contiguity, between 0.947 and 0.952. The lowest line under rook contiguity, which the legend does not mark, is Checkerboard, for which τ is not identified, and its estimates are kept and flagged. The five values of τ share their assignments and noise.



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure S11: Coverage of the 95% confidence intervals by design across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, pooled over the five incidence configurations. Each point averages the 160 scenarios of one design at one value of τ. The dashed red line marks the nominal 0.95 level, and the gray dashed line marks the mean coverage of simple random sampling.

10 Rook contiguity

Under rook contiguity at $\tau = 1$, pooled over the five incidence configurations, the mean-MSE order was Incidence-Guided Saturation (0.072), Balanced Quartiles (0.084), 2×2 Blocking (0.128), Isolation Buffer (0.133), High Incidence Focus (0.206), and Checkerboard (0.480) (Table S7; Figure S12), and the same order held within each of the five incidence configurations (Table S8). Incidence-Guided Saturation had the lowest MSE in 97 of the 160 rook blocks (61%) and Balanced Quartiles in 60 (38%). Coverage was between 0.948 and 0.952 for every design except Checkerboard, whose coverage was 0.162 (Figure S13). Within single incidence configurations, the corresponding ranges were 0.944 to 0.954, and 0.15 to 0.16 for Checkerboard. With SRS as a seventh member, the Friedman test rejected equal ranks ($\chi^2 = 795.77$, df $= 6$, 160 blocks, $p < 2.2 \times 10^{-16}$). Nemenyi comparisons separated 17 of 21 pairs; the four they did not separate were Incidence-Guided Saturation against Balanced Quartiles ($p = 0.66$) and SRS ($p = 0.14$), Balanced Quartiles against SRS ($p = 0.97$), and 2×2 Blocking against Isolation Buffer ($p = 1.00$). Holm-adjusted Wilcoxon tests separated every pair except 2×2 Blocking against Isolation Buffer and Balanced Quartiles against SRS (both $p = 0.21$). In the surface-level comparison, Incidence-Guided Saturation had a lower MSE than Balanced Quartiles in 48 of the 50 units (mean difference -0.012; -0.014, -0.010;

Wilcoxon $p = 9.9 \times 10^{-10}$). 2×2 Blocking and Isolation Buffer were close: 2×2 Blocking had the lower MSE in 36 of the 50 units (72%; mean difference -0.005; -0.008, -0.002; Wilcoxon $p = 0.0027$), but Isolation Buffer had the slightly better average rank (4.21 against 4.33), and the two designs traded places in mean MSE at $\tau = 3$ (see Section 9). Every other adjacent pair differed in all 50 units.

Checkerboard under rook contiguity is a different case: τ is not identified, and the coefficient on $\mathbf{Z}$ estimates $\tau - \gamma$ (Section 3.1). The estimates behaved accordingly. At $\tau = 1$, Checkerboard's mean bias was -0.502, -0.600, -0.704, and -0.802 at $\gamma = 0.5$, 0.6, 0.7, and 0.8, while its mean standard deviation stayed between 0.20 and 0.21; squared bias, averaged over scenarios, accounted for 0.438 of its MSE of 0.480 (Figure S14 shows the decomposition for the Poisson configuration with $\rho_X = 0.20$). Its coverage fell from 0.337 at $\gamma = 0.5$ to 0.031 at $\gamma = 0.8$. Because $\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$, the two spillover covariates in Equation S2 coincide, $\gamma\,\mathbf{WZ} \circ (\mathbf{1} - \mathbf{Z}) = \gamma(\mathbf{1} - \mathbf{Z}) = \gamma\,\mathbf{WZ}$, so the two regimes define the same outcome model and the same expected MSE; the observed values, 0.477 under the both-arms regime and 0.483 under the control-only regime, differ only by Monte Carlo error, because the two regimes use separate noise draws. Under queen contiguity, by contrast, Checkerboard's coverage stayed between 0.949 and 0.952 at every level of ρ; under rook contiguity it was between 0.146 and 0.171 (Table S13). Rook Checkerboard estimates were precise but centered on the wrong quantity, the opposite of its imprecise but valid queen estimates. We keep these estimates in every ranking and flag them.

Table S13: Checkerboard's 95% confidence interval coverage across the four levels of outcome spatial dependence ρ at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of $\gamma \times$ 2 spillover regimes). Under rook contiguity, τ is not identified for Checkerboard. Its estimates are kept and flagged. The nominal level is 0.95.

Neighbor type	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Queen	0.950	0.952	0.949	0.949
Rook	0.171	0.168	0.161	0.146

Figures S12, S13 and S14 are the rook-contiguity versions of three queen-contiguity figures in Section 7. Each has the layout of its queen counterpart, with the designs reordered for rook contiguity. Under rook contiguity, τ is not identified for Checkerboard ($\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$), and its estimates are kept and flagged. Figure S12 corresponds to Figure S6.

Figure S13 corresponds to Figure S7. Every design except Checkerboard has coverage close to the nominal level, whereas Checkerboard's coverage, 0.162 pooled over the five incidence configurations (Table S7), is far below it.

Figure S14 corresponds to Figure S8. Nearly all of Checkerboard's MSE is squared bias, because its coefficient on $\mathbf{Z}$ estimates $\tau - \gamma$ (Section 3.1).

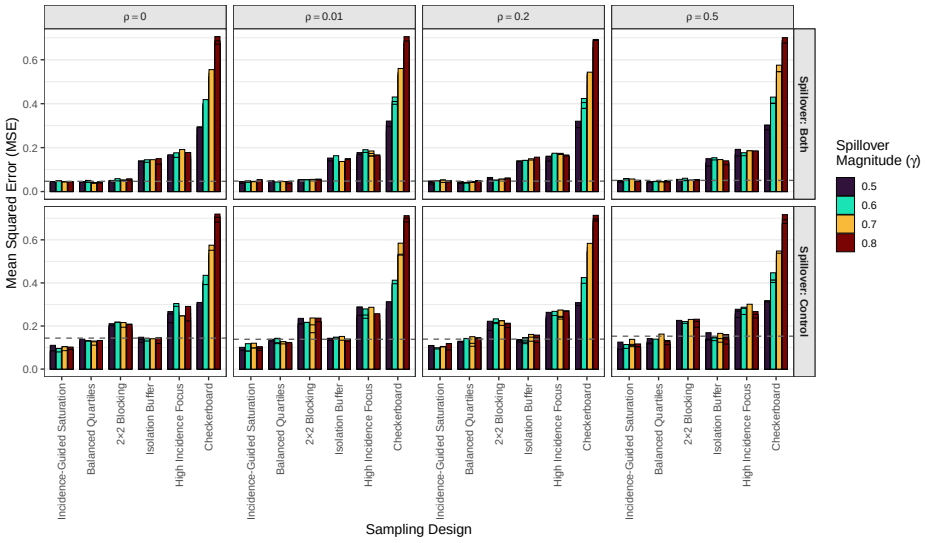


Figure S12: Mean squared error of the oracle estimate of τ by design at $\tau = 1$, under rook contiguity; layout as in Figure S6. Checkerboard is flagged (τ not identified).

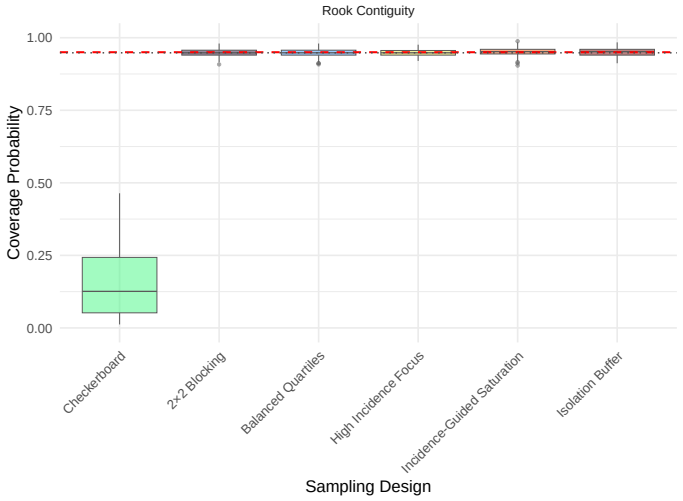


Figure S13: Coverage of the 95% confidence intervals by design at $\tau = 1$, under rook contiguity; layout as in Figure S7. Checkerboard is flagged (τ not identified).

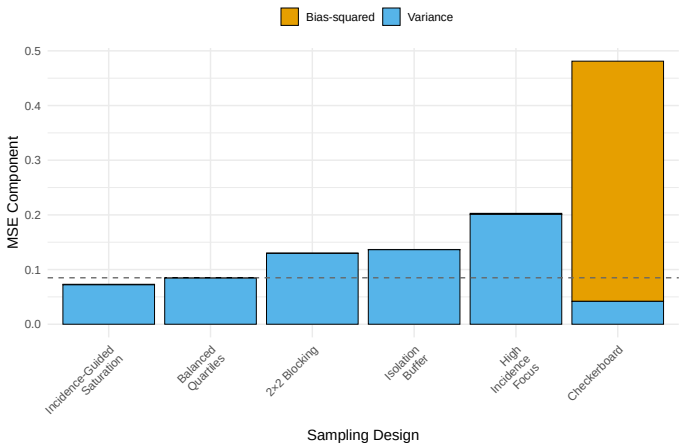


Figure S14: Squared bias and variance of $\hat{\tau}$ by design at $\tau = 1$ for the Poisson configuration with $\rho_X = 0.20$, under rook contiguity; layout as in Figure S8. Checkerboard is flagged (τ not identified).

11 Non-oracle sensitivity analysis

The non-oracle model, which omits the spillover term, was biased for every design (Table S14 gives both models' MSE, bias, and coverage for each design). At $\tau = 1$, and setting aside Checkerboard under rook contiguity, its bias ranged from -0.12 to -0.40 and its coverage from 0.54 to 0.91 across designs and neighbor types. Under queen contiguity its bias ranged from -0.123 (Incidence-Guided Saturation) to -0.244 (2×2 Blocking) and its coverage from 0.727 (2×2 Blocking) to 0.907 (High Incidence Focus); under rook contiguity the most biased design other than Checkerboard was Isolation Buffer (bias -0.400, coverage 0.536). For Checkerboard under rook contiguity, the non-oracle estimates were identical to the oracle estimates (MSE 0.480, bias -0.652, coverage 0.162), because once the aliased spillover column is dropped the two models are the same.

Omitting the spillover term also removes a column that can be nearly collinear with $\mathbf{Z}$, and the non-oracle estimates were accordingly less variable for some designs. Under queen contiguity, the non-oracle MSE was lower than the oracle MSE for every design, most of all for Checkerboard (oracle 1.087, non-oracle 0.108) and 2×2 Blocking (oracle 0.319, non-oracle 0.121). Under rook contiguity it was higher for every design except 2×2 Blocking (0.128 under both models) and Checkerboard (identical). These results do not change the recommendation. Incidence-Guided Saturation also had the lowest non-oracle MSE under both neighbor types (0.087 under queen contiguity and 0.086 under rook contiguity), and where a design's non-oracle MSE was lower, the estimate was biased and its intervals under-covered, so the lower MSE does not support valid inference.

Table S14 gives the mean squared error, bias, and coverage of the oracle and non-oracle models for each of the six designs, as summarized above.

Table S14: Oracle and non-oracle estimates of τ by design at $\tau = 1$, under queen and rook contiguity, pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type. The non-oracle model omits Spill. Rows follow oracle MSE under queen contiguity. Under rook contiguity, τ is not identified for Checkerboard ($\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$). Its estimates are kept and flagged, and the two models coincide.

	Oracle			Non-oracle		
Design	MSE	Bias	Coverage	MSE	Bias	Coverage
Queen contiguity (primary)						
Incidence-Guided Saturation	0.091	0.032	0.947	0.087	−0.123	0.825
Balanced Quartiles	0.131	0.017	0.946	0.098	−0.176	0.786
Isolation Buffer	0.160	0.003	0.949	0.091	−0.175	0.892
High Incidence Focus	0.245	0.031	0.948	0.219	−0.180	0.907
2×2 Blocking	0.319	0.010	0.948	0.121	−0.244	0.727
Checkerboard	1.087	0.001	0.950	0.108	−0.206	0.758
Rook contiguity (sensitivity)						
Incidence-Guided Saturation	0.072	0.018	0.952	0.086	−0.121	0.825
Balanced Quartiles	0.084	0.008	0.948	0.097	−0.176	0.789
Isolation Buffer	0.133	0.003	0.950	0.211	−0.400	0.536
High Incidence Focus	0.206	0.018	0.948	0.223	−0.192	0.903
2×2 Blocking	0.128	0.004	0.948	0.128	−0.257	0.710
Checkerboard (flagged)	0.480	−0.652	0.162	0.480	−0.652	0.162

12 Robustness, the SRS benchmark and allocation-specific risk

12.1 Robustness across blocks and the recommended design

Table S15 gives the distribution of MSE across the 160 queen blocks at $\tau = 1$. Incidence-Guided Saturation had the lowest median (0.080) and the lowest worst-case MSE of any design (0.169). Isolation Buffer had the narrowest range (0.128 to 0.206), because its MSE does not depend on the spillover regime, but at a higher level. Balanced Quartiles had a low median (0.115) and a worst case of 0.290. The distributions of 2×2 Blocking and Checkerboard reflected their dependence on the spillover regime: both had low first quartiles (0.062 and 0.063) and high upper quartiles (0.580 and 2.093). Checkerboard had the highest MSE in all 80 control-only blocks; in the 80 both-arms blocks, the highest MSE belonged to High Incidence Focus in 42 and to Isolation Buffer in 38. Under rook contiguity, Incidence-Guided Saturation also had the lowest worst-case MSE (0.138).

Table S15: Distribution of block-level mean squared error by design across the 160 blocks under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (32 blocks in each). At a fixed value of τ, each block contributes one scenario per design. Rows are sorted by median, which here coincides with the order of pooled mean MSE under queen contiguity (Table S7).

Design	Best	Q25	Median	Q75	Worst
Incidence-Guided Saturation	0.037	0.045	0.080	0.135	0.169
Balanced Quartiles	0.034	0.042	0.115	0.221	0.290
Isolation Buffer	0.128	0.150	0.159	0.169	0.206
High Incidence Focus	0.139	0.160	0.234	0.322	0.424
2×2 Blocking	0.048	0.062	0.271	0.580	0.719
Checkerboard	0.051	0.063	0.899	2.093	2.843

Among the six designs in the detailed comparison, Incidence-Guided Saturation combines low pooled MSE with comparatively favorable control-only performance. The SRS comparison in Section 12.2 qualifies this. A saturation design is a reasonable choice when control-only spillover is plausible, and SRS remains competitive under both-arms spillover. Saturation Quadrants performed similarly, so incidence guidance lets a trial direct its highest treatment saturations to the highest-incidence regions with little or no loss of estimation accuracy. The scenario-level worst cases in Table S15 summarize variation across parameter settings, which is different from the conditional risk across assignments within one setting (Section 12.3).

12.2 Comparison with SRS

At $\tau = 1$ under queen contiguity, pooled over the five incidence configurations and all settings of ρ, γ and spillover regime, Incidence-Guided Saturation has mean MSE 0.091 against 0.128 for SRS, and Saturation Quadrants has 0.090. Table S16 separates the regimes and includes all eight strategies. Incidence-Guided Saturation's control-only ratio to SRS is 0.631, whereas its both-arms ratio is 1.083. SRS, Balanced Quartiles and Balanced Halves have similar both-arms accuracy. A small mean difference does not show equivalence, and pooling the regimes would hide that the two regimes favor different designs.

Figure S15 tests each design against SRS within each regime, over the 50 independent configuration-by-surface units. Under control-only spillover, Incidence-Guided Saturation had 36.9% lower MSE than SRS (95% interval 34.5% to 39.2% lower), Saturation Quadrants 36.7% lower and Isolation Buffer 25.7% lower, each in all 50 units (Holm-adjusted $p = 6.2 \times 10^{-9}$). Under

both-arms spillover, Incidence-Guided Saturation had 8.3% higher MSE than SRS (5.6% to 11.0% higher; $p = 6.1 \times 10^{-6}$) and Saturation Quadrants 4.3% higher ($p = 0.017$). Balanced Quartiles and Balanced Halves did not differ significantly from SRS under either regime, and the remaining designs had significantly higher MSE than SRS under both-arms spillover, as did High Incidence Focus, 2×2 Blocking and Checkerboard under control-only spillover.

Under rook contiguity the saturation designs had 23% to 24% lower control-only MSE than SRS and 3% to 5% higher both-arms MSE, significant for Incidence-Guided Saturation ($p = 0.0075$) but not for Saturation Quadrants ($p = 0.15$), and Isolation Buffer had 5.8% higher control-only MSE than SRS.

The mean is only part of the comparison, because a trial realizes a single allocation. Under control-only spillover, the worst 10% of SRS allocations on the grid had mean conditional MSE 60% above that of the average SRS allocation (0.346 against 0.217), and the saturation designs' worst-decile MSE was 0.54 to 0.55 times SRS's on the grid and 0.64 to 0.68 times in NC. Under both-arms spillover, their worst-decile MSE was 1.06 to 1.07 times SRS's on the grid and 0.93 to 0.94 times in NC (Table S19), so the small mean disadvantage there comes with no worse downside in NC.

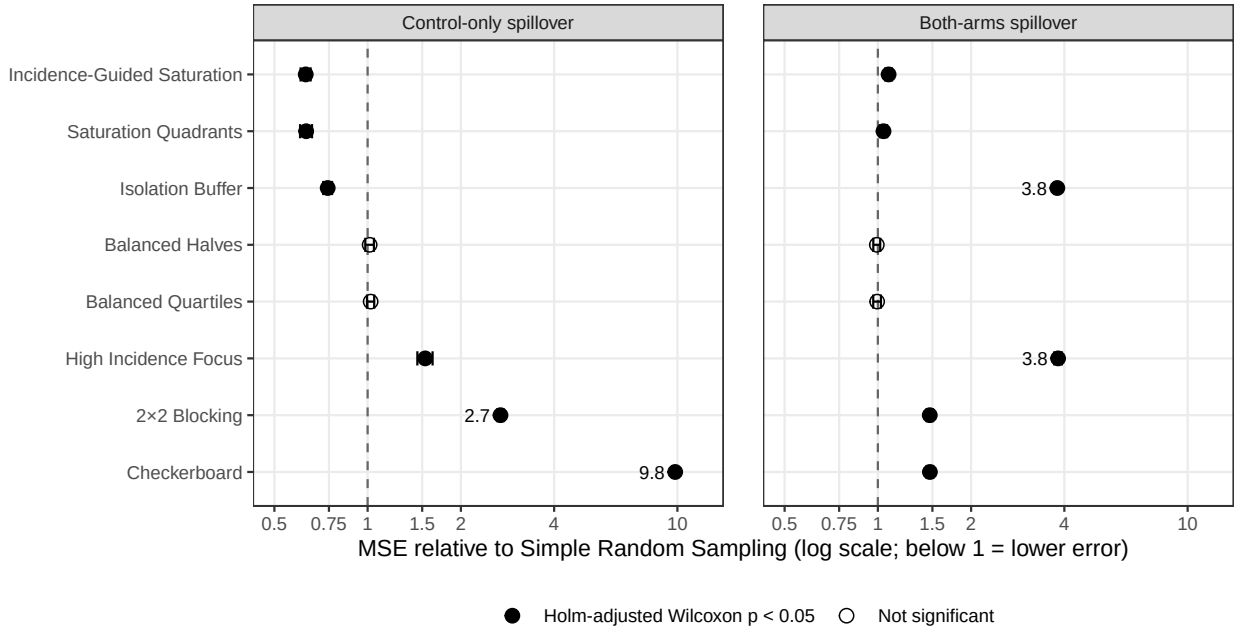


Figure S15: Each design's mean squared error (MSE) relative to simple random sampling (SRS) at $\tau = 1$ under queen contiguity, within each spillover regime. Each point is one plus the mean paired MSE difference over the 50 configuration-by-surface units divided by SRS's mean MSE, with its t_{49} 95% interval; 1 marks equal error. Filled points are significant after Holm's adjustment over the eight comparisons with SRS (Wilcoxon signed-rank test). The axis is logarithmic, ratios above 2 are printed, and rows are sorted by the control-only ratio.

12.3 Allocation-specific risk

Averaged over the 200 units per regime (fixed surfaces crossed with the extreme combinations of ρ and γ), Incidence-Guided Saturation's control-only mean MSE is 0.140 against SRS 0.217, the

Table S16: Full grid benchmark at $\tau = 1$, queen contiguity. Each regime-specific entry averages 80 scenarios (five incidence configurations, four ρ and four γ values). MSE and coverage use the oracle fit; coverage is that of 95% t-based intervals with a degrees-of-freedom-corrected standard error. Simple random sampling treats 50 of the 100 clusters at random. No value pools queen and rook.

	Both arms		Control only	
Design	MSE	Coverage	MSE	Coverage
Checkerboard	0.062	0.948	2.111	0.951
High Incidence Focus	0.162	0.950	0.329	0.945
Saturation Quadrants	0.044	0.952	0.136	0.940
Isolation Buffer	0.161	0.948	0.159	0.950
2×2 Blocking	0.062	0.947	0.576	0.950
Balanced Quartiles	0.042	0.949	0.219	0.942
Balanced Halves	0.042	0.949	0.218	0.943
Incidence-Guided Saturation	0.046	0.949	0.135	0.945
Simple Random Sampling	0.042	0.950	0.215	0.946

estimated 90th percentile of allocation-specific MSE is 0.173 against 0.296, and the estimated worst-decile mean is 0.191 against 0.346, lower than SRS in all 200 units on both measures. Under both arms, the corresponding mean MSE is 0.046 against 0.043 and worst-decile mean 0.058 against 0.054. Saturation Quadrants also performs favorably under control-only spillover. Table S19 (Section 14) compares every design with SRS on both criteria. These results come from the extreme parameter settings only and give qualified support to saturation on downside risk. The refined setting kept the control-only direction but showed that outcome noise accounts for much of the estimated variation between allocations, especially under both arms. Quantiles and worst-decile membership estimated from a finite number of outcomes remain uncertain. We therefore treat the NC application (Sections 13 and 14) as a separate, setting-specific assessment rather than extrapolate exact tail protection from these settings.

13 Application: accuracy, inference and reach

The main run generated 7,598,000 independent outcome fits. Every setting is complete and meets its prespecified precision targets for mean MSE and coverage. The largest relative Monte Carlo standard error of mean MSE is 0.049 and the largest Monte Carlo standard error of coverage is 0.00892. There are no warnings, aliased columns, estimates of ρ at the boundary of its parameter space or estimation failures. The tail refinement adds 2,280,000 independent outcomes, and counting the outcomes reused from the main run only once gives 9,878,000 distinct outcomes across the two stages.

Figure S16 compares the eight designs with SRS by year and regime. Setting-specific comparisons are in Figure S18. Under control-only spillover, Saturation Quadrants has descriptive mean MSE 0.223 against SRS 0.308, a 27.4% reduction. Incidence-Guided Saturation reduces mean MSE by 21.1–29.9% across the four observed years (Table S17). Both saturation designs have lower mean MSE than SRS in every primary parameter setting, and approximate Monte Carlo intervals for each setting-specific difference excluded zero. Neither is uniquely best. The two performed similarly, so incidence guidance lets a trial direct its highest treatment saturations to the highest-incidence regions with little or no loss of estimation accuracy, and their advantage over SRS comes from the

saturation structure both designs share. High Incidence Focus is year-sensitive under control-only spillover and cannot be recommended uniformly.

Under both-arms spillover, SRS mean MSE is 0.076, Saturation Quadrants 0.075, and Incidence-Guided Saturation ranges from 0.074 to 0.077 across years. SRS remains competitive with balancing, blocking and saturation. Small point differences do not establish equivalence. High Incidence Focus has low both-arms MSE in the primary model, but its allocation is deterministic for these fixed rates. That is model-based accuracy conditional on one allocation, not evidence of randomization-based causal validity.

Mean coverage of the 95% intervals across the four years was between 94.1% and 94.9% for every design and regime, and between 94.1% and 95.1% in single years. The Wald interval covered only 92.6% to 93.6% (Table S24). Meeting the Monte Carlo precision targets did not repair that finite-sample undercoverage, but the corrected t interval described in Section 5.3 did (Section 15.2). Table S18 gives bias and coverage for every design, year and regime. Checkerboard's treatment effect is identified on this irregular graph, but weak separation of treatment from spillover makes its estimates imprecise. This differs from the exact non-identification under rook contiguity on the rectangular grid.

Population shares put these results in context. High Incidence Focus treats 29 clusters but only about 22–24% of working-age population, while Isolation Buffer treats a mean of 15.89 clusters and about 27%. These population denominators measure geographic reach, not the number of healthcare professionals educated, intervention cost or participant-level precision. 2×2 Blocking treats 28, 31, 27 and 29 clusters by year. Table S5 and Figure S17 give the complete budgets and shares.

Incidence-Guided Saturation sends more treatment to high-incidence regions when incidence differs between regions. On the NC map it differs little, because the four regions' mean incidence ranks lie between 24 and 34 of 58 in every year. With all 58 colleges, Incidence-Guided Saturation's treated colleges held on average 51.1% of the summed college incidence rates, against 50.4% for Saturation Quadrants, 50.0% for SRS and 62.2% for High Incidence Focus. Shares of deaths run the other way (49.2%, 49.6%, 49.7% and 35.4%), because the highest-rate colleges serve small populations. On the grid, where regional incidence differed more and equal cluster populations make rate share and death share coincide, incidence guidance raised the treated share of incidence over Saturation Quadrants by 2.1 to 3.1 percentage points on average across the five incidence configurations, while High Incidence Focus reached 0.75 to 0.77 by giving up randomization over the whole map. Across surfaces the gain rose almost linearly with the spread of the regions' mean incidence ranks (correlation 0.84 to 0.99), and the NC regions lie at the low end of that spread (Figure S20).

Table S17: Primary application yearly mean MSE, queen contiguity; equal weights over the six ρ/γ pairs. Saturation Quadrants and SRS reuse the same simulations across years.

	Both arms		Control only		
Year	SRS	Incidence-Guided Saturation	SRS	Saturation Quadrants	Incidence-Guided Saturation
2018	0.076	0.074	0.308	0.223	0.243
2019	0.076	0.076	0.308	0.223	0.216
2020	0.076	0.077	0.308	0.223	0.220
2021	0.076	0.075	0.308	0.223	0.237

Table S18 and Figure S19 give MSE, bias and coverage for every design in every year, and Table S5 gives the budgets. More granular setting-specific results, precision components, residual

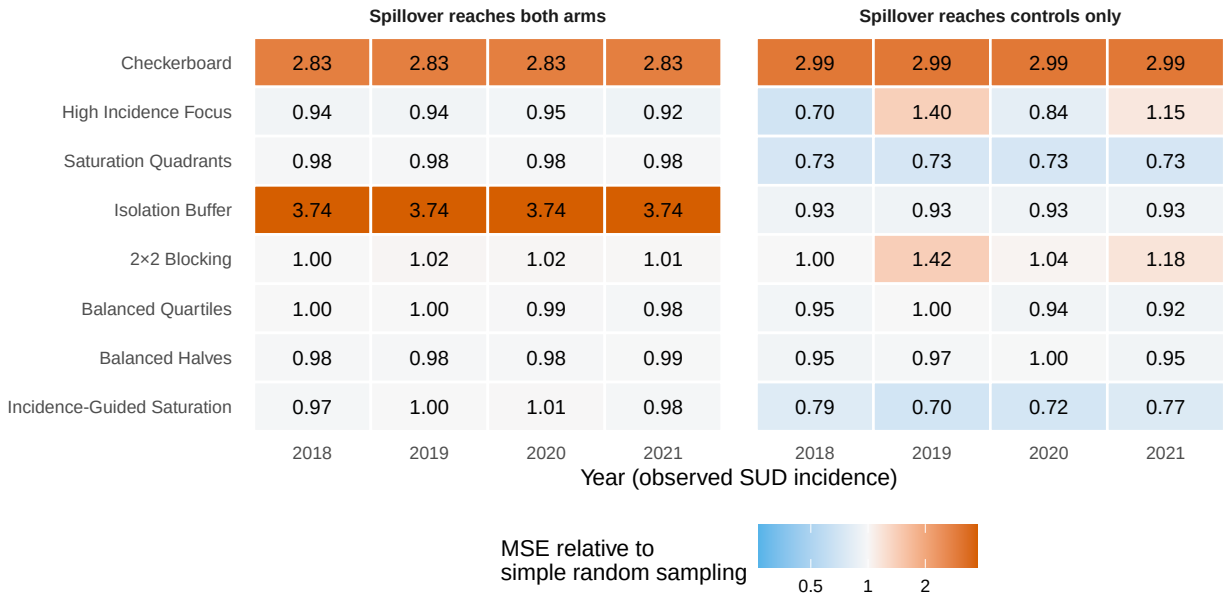


Figure S16: Completed primary application: yearly mean MSE divided by year- and regime-matched SRS MSE for the eight other designs, with queen weights and education outcomes. Each cell averages six ρ/γ settings equally before taking the ratio. Blue marks lower error than SRS and orange higher; colors use a logarithmic ratio scale and cells print the ratios. Years that share simulations are not independent replications.

treatment information and warning and failure records are kept as aggregate output tables. These records were checked, so the absence of warnings and failures is a verified result.

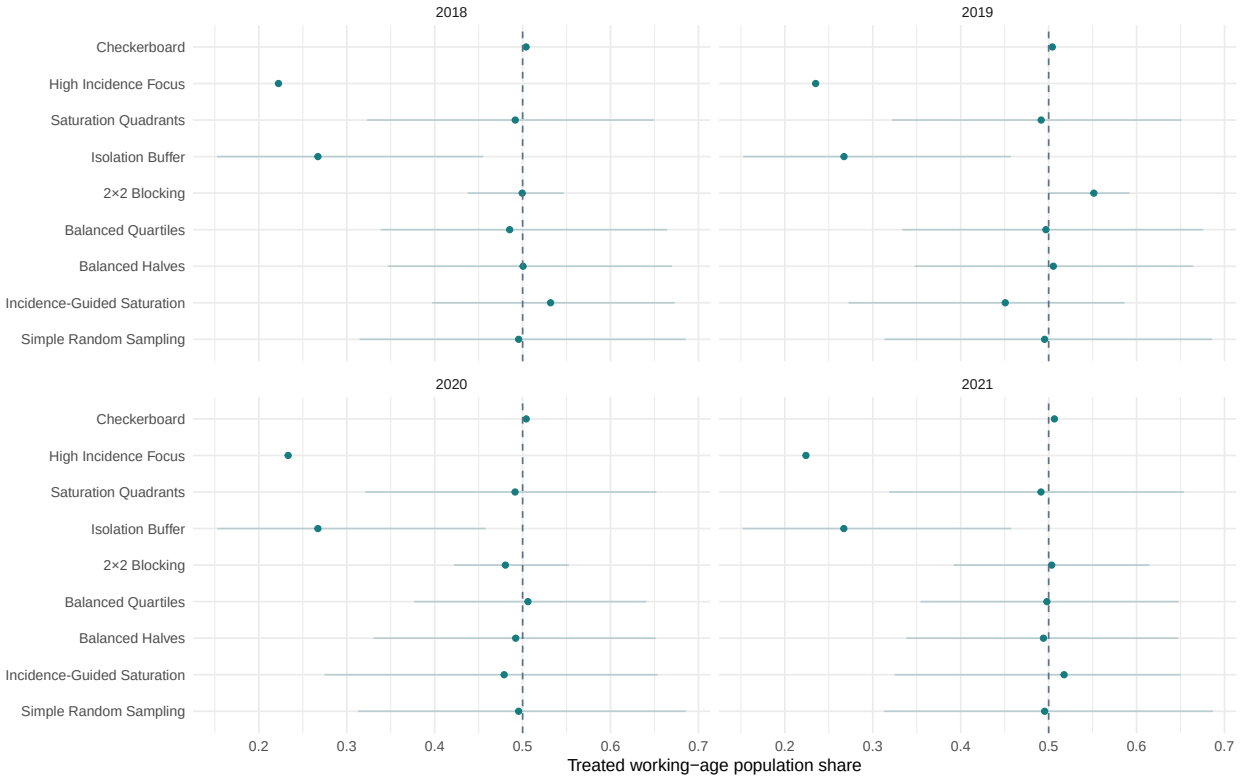


Figure S17: Annual population shares treated under primary queen assignments, using the both-arms runs (the set of possible assignments does not depend on the regime). Points mark the mean and lines the range over sampled allocations. Designs whose assignments do not depend on annual incidence share performance across years, but their annual population shares are recalculated. Population aged 18–64 is geographic reach rather than education-participant enrollment.

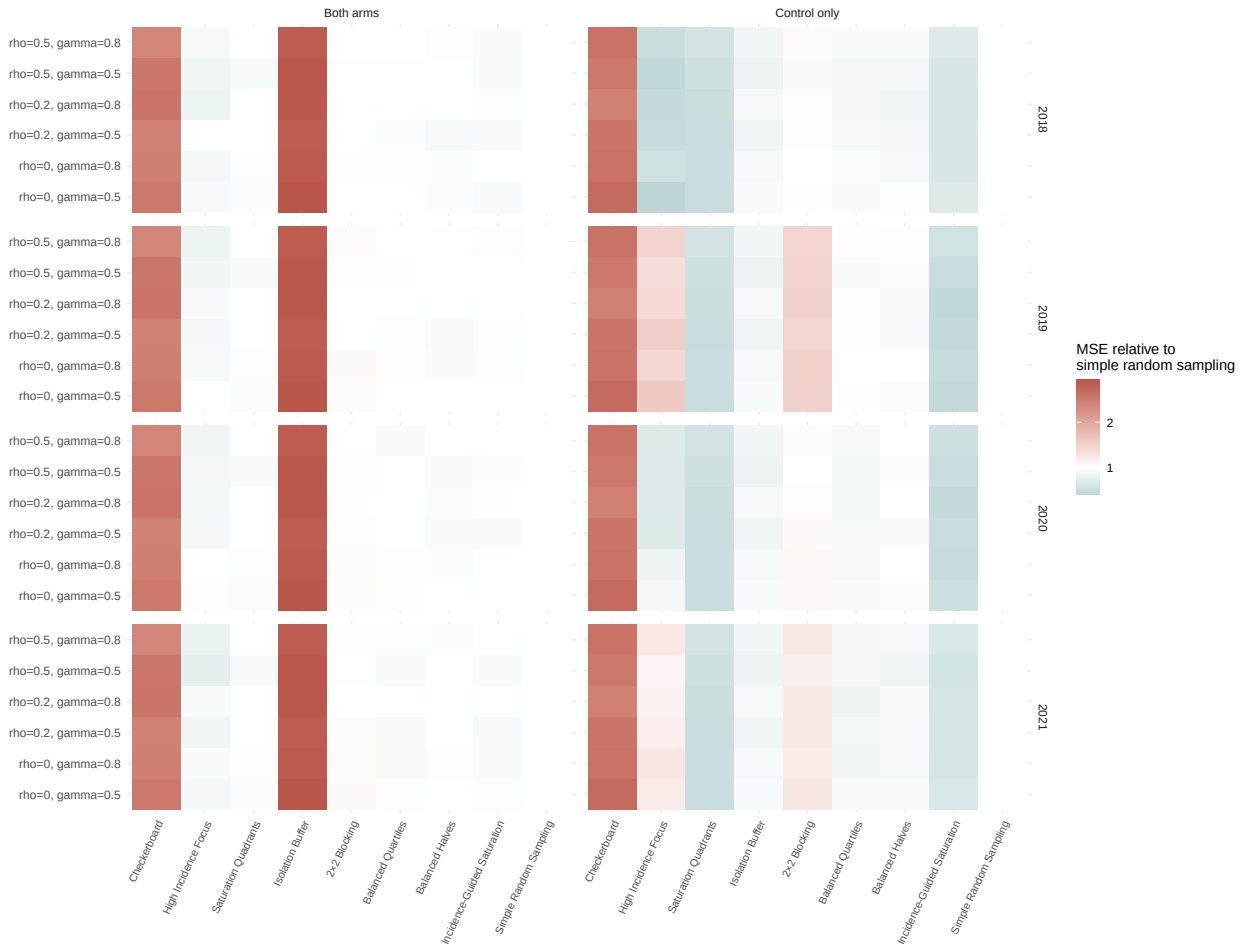


Figure S18: Primary queen setting-specific mean MSE ratios to year-matched SRS. Teal cells have lower MSE than SRS and red cells higher; setting-specific Monte Carlo errors are in the source table. Parameters and spillover regimes are kept separate; this plot does not pool neighbor definitions.

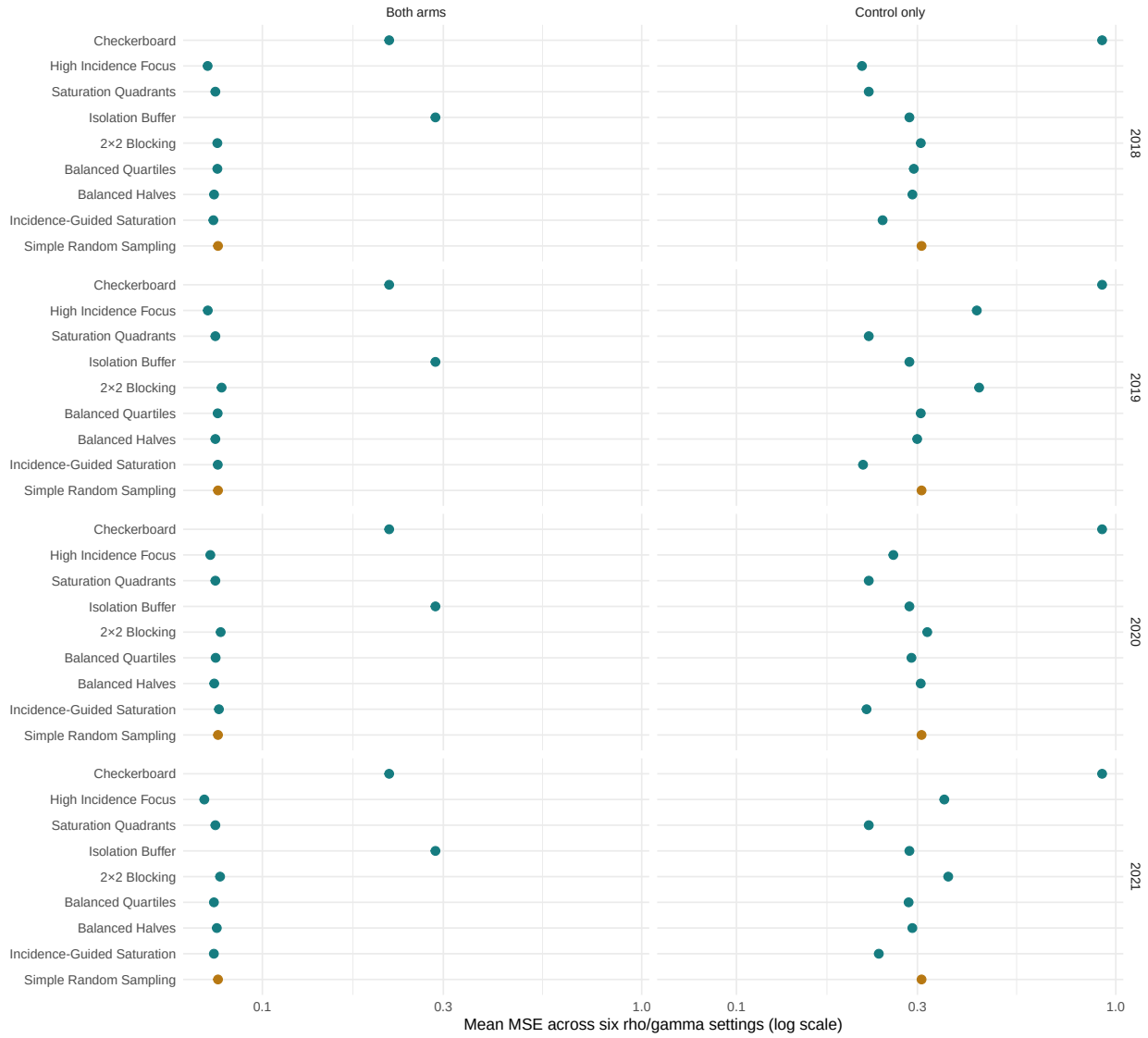


Figure S19: Completed primary application: mean MSE by design, planning year and spillover regime, queen contiguity and education outcome. Each entry equally averages the six ρ/γ pairs, and simple random sampling is shown in gold. Both regime columns use a common logarithmic MSE scale. Years that share the same simulations are repeated reporting, not independent replications; pooled Monte Carlo intervals are not supplied.

Table S18: Primary queen application performance by design and planning year. Each entry averages the six ρ/γ pairs of one spillover regime with equal weight; coverage is that of the corrected t interval. Checkerboard, Saturation Quadrants, Isolation Buffer and SRS use the same simulations in every year, because annual incidence enters neither their allocation nor the outcome model, so their rows repeat and are not independent annual confirmation.

		Both arms			Control only		
Design	Year	MSE	Bias	Coverage	MSE	Bias	Coverage
Checkerboard	2018	0.216	-0.007	0.949	0.920	0.010	0.948
	2019	0.216	-0.007	0.949	0.920	0.010	0.948
	2020	0.216	-0.007	0.949	0.920	0.010	0.948
	2021	0.216	-0.007	0.949	0.920	0.010	0.948
High Incidence Focus	2018	0.072	0.016	0.948	0.214	0.024	0.948
	2019	0.072	0.006	0.951	0.430	0.023	0.945
	2020	0.073	0.009	0.946	0.259	0.033	0.946
	2021	0.070	0.010	0.951	0.353	0.030	0.945
Saturation Quadrants	2018	0.075	0.021	0.949	0.223	0.046	0.941
	2019	0.075	0.021	0.949	0.223	0.046	0.941
	2020	0.075	0.021	0.949	0.223	0.046	0.941
	2021	0.075	0.021	0.949	0.223	0.046	0.941
Isolation Buffer	2018	0.286	0.009	0.947	0.286	0.011	0.947
	2019	0.286	0.009	0.947	0.286	0.011	0.947
	2020	0.286	0.009	0.947	0.286	0.011	0.947
	2021	0.286	0.009	0.947	0.286	0.011	0.947
2×2 Blocking	2018	0.076	0.010	0.948	0.306	0.028	0.944
	2019	0.078	0.014	0.945	0.436	0.031	0.942
	2020	0.078	0.012	0.949	0.318	0.028	0.945
	2021	0.077	0.011	0.947	0.362	0.029	0.943
Balanced Quartiles	2018	0.076	0.017	0.947	0.294	0.031	0.944
	2019	0.076	0.017	0.948	0.306	0.037	0.944
	2020	0.075	0.016	0.947	0.289	0.036	0.943
	2021	0.075	0.018	0.948	0.284	0.035	0.944
Balanced Halves	2018	0.075	0.016	0.948	0.291	0.040	0.945
	2019	0.075	0.017	0.949	0.300	0.036	0.945
	2020	0.075	0.015	0.948	0.306	0.037	0.944
	2021	0.076	0.016	0.946	0.291	0.034	0.944
Incidence-Guided Saturation	2018	0.074	0.017	0.949	0.243	0.046	0.943
	2019	0.076	0.023	0.949	0.216	0.052	0.941
	2020	0.077	0.021	0.947	0.220	0.047	0.942
	2021	0.075	0.020	0.950	0.237	0.045	0.942
Simple Random Sampling	2018	0.076	0.013	0.947	0.308	0.038	0.945
	2019	0.076	0.013	0.947	0.308	0.038	0.945
	2020	0.076	0.013	0.947	0.308	0.038	0.945
	2021	0.076	0.013	0.947	0.308	0.038	0.945

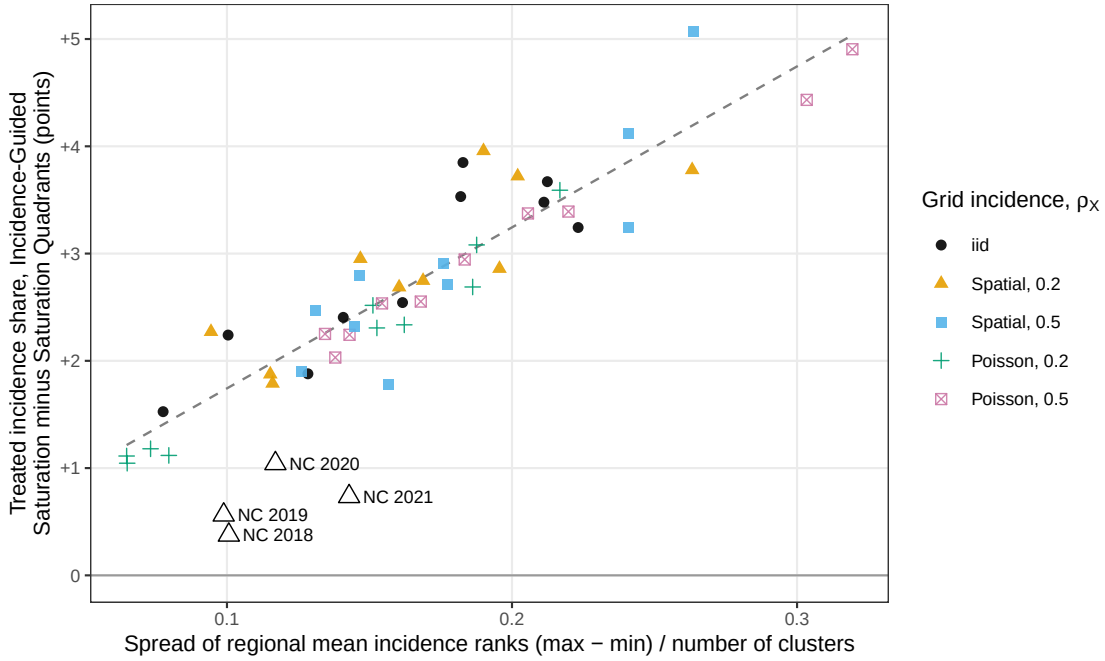


Figure S20: Gain in treated incidence share from incidence guidance (Incidence-Guided Saturation minus Saturation Quadrants) against the spread of the regions' mean incidence ranks, the largest minus the smallest regional mean rank divided by the number of clusters. Filled symbols are the 50 grid surfaces (queen, production allocations) by incidence configuration, with a least-squares line fitted to them. Open triangles are the fixed NC regions in each of the four planning years, whose spread divides by 58 clusters rather than 100.

14 Application: allocation downside

At the two refined queen settings, $\rho = 0$ and $\rho = 0.5$ with $\gamma = 0.8$, under control-only spillover, Saturation Quadrants' estimated 90th percentile of allocation-specific MSE is 31.8–32.9% below SRS and its estimated worst-decile mean 34.6–37.8% below. Incidence-Guided Saturation's corresponding reductions are 20.8–38.6% and 20.7–43.4%. The pattern persists when the worst decile is identified and evaluated on separate halves of the simulated outcomes (Section 5.4). Figure S22 shows the $\rho = 0.5$ setting. The tail estimates rest on a finite number of outcomes per allocation, so worst-decile membership is uncertain. They describe the spread of conditional MSE across the sampled allocations, not quantiles of realized errors, and because the refinement reused the main run's allocations they are not a fresh allocation sample. Read this way, the results show a descriptive downside benefit alongside lower average error, conditional on the studied model and sampled assignments.

Table S19 compares every design with SRS on both criteria, mean conditional MSE and the mean of the worst 10% of sampled allocations, on the grid and at the refined NC settings. Under control-only spillover, both saturation designs had lower mean and worst-decile MSE than SRS in both settings. Isolation Buffer also had a lower worst decile than SRS under control-only spillover (0.62 of SRS on the grid and 0.91 in NC), but under both-arms spillover its mean was 3.80 and 3.69 times that of SRS, and it treated fewer clusters. High Incidence Focus has a single allocation in NC. That allocation's MSE was below the SRS worst-decile mean under both regimes (0.76 under both arms and 0.64 under control-only spillover), with mean MSE close to SRS (0.93 and 1.03). On the grid, however, its both-arms mean was 3.81 times that of SRS, and its NC result depends on the incidence model (see the matched baseline sensitivity below). Balanced Quartiles and Balanced Halves matched SRS on both criteria. Checkerboard and 2×2 Blocking did not improve on SRS on either criterion in any setting.

Table S19: Mean conditional MSE and worst-decile mean conditional MSE of each design divided by those of SRS in the same setting and spillover regime (below 1 is lower error than SRS). Grid: allocation-risk study, queen contiguity, $\tau = 1$, averaged over 200 units, each a fixed incidence surface with one extreme combination of ρ and γ. NC: refined queen settings $\rho \in \{0, 0.5\}$, $\gamma = 0.8$, averaged equally over 2018–2021 and both values of ρ, with the map adaptations of the designs. A design with one allocation has worst-decile mean equal to its mean.

	Grid				NC application			
	Both arms		Control only		Both arms		Control only	
Design	Mean	Worst 10%	Mean	Worst 10%	Mean	Worst 10%	Mean	Worst 10%
Checkerboard	1.41	1.11	9.94	6.22	2.68	2.20	3.02	1.88
High Incidence Focus	3.81	3.05	1.51	0.96	0.93	0.76	1.03	0.64
Saturation Quadrants	1.06	1.06	0.62	0.54	0.99	0.94	0.73	0.64
Isolation Buffer	3.80	3.97	0.75	0.62	3.69	4.75	0.93	0.91
2×2 Blocking	1.44	1.57	2.69	2.64	1.00	1.02	1.15	1.18
Balanced Quartiles	0.99	0.99	1.00	1.01	0.98	0.96	0.95	1.01
Balanced Halves	0.99	0.99	1.01	1.01	0.98	0.96	0.97	0.99
Incidence-Guided Saturation	1.07	1.07	0.65	0.55	0.98	0.93	0.75	0.68
Simple Random Sampling	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Figure S21 shows the full allocation-to-allocation distribution for the six designs and SRS. For each allocation in Figure S21, MSE is estimated from repeated simulated outcomes (100 in the simulation study, at least 400 in the application study). Percentiles are computed within each setting and then averaged, so the bars show allocation-to-allocation spread, and outcome Monte

Carlo noise in the per-allocation estimates widens them slightly. Under control-only spillover, most Incidence-Guided Saturation allocations had lower MSE than the average SRS allocation in both studies, whereas the SRS distribution extended well above its mean.

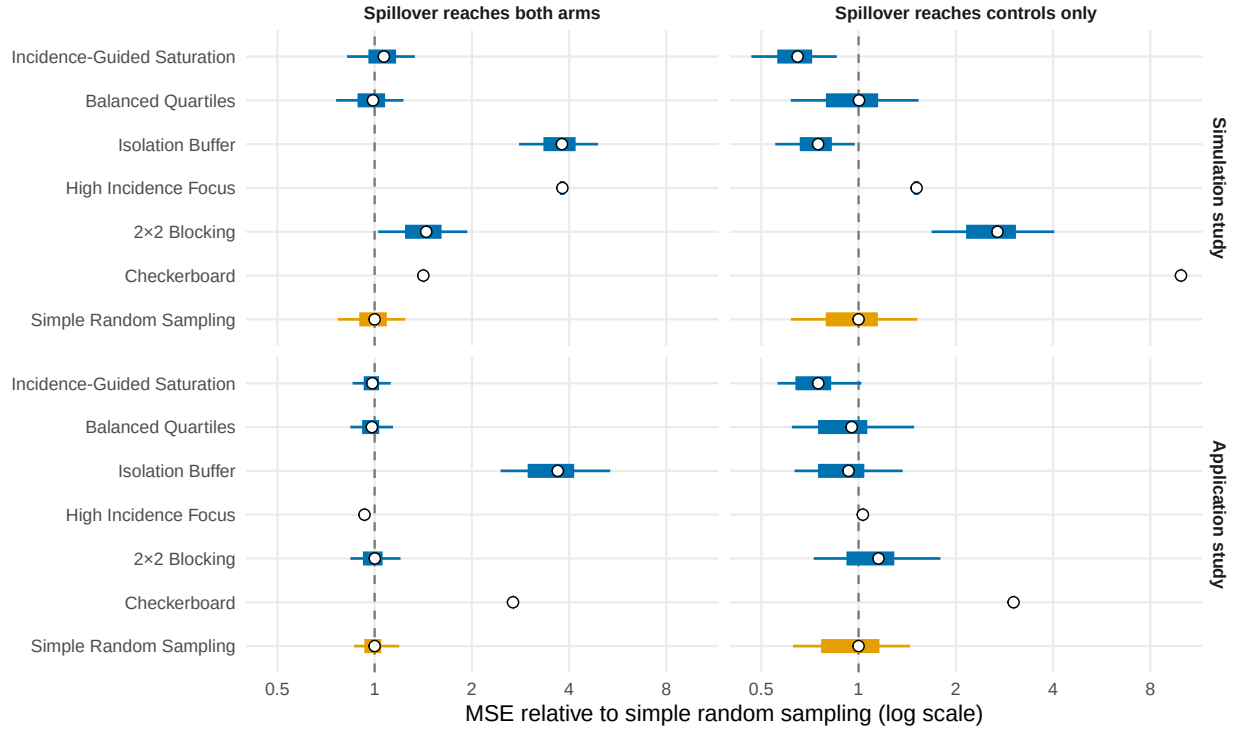


Figure S21: Distribution of allocation-specific MSE for the six designs and simple random sampling (SRS), relative to the mean MSE of SRS in the same setting (queen weights). Simulation study: grid allocation-risk study at $\tau = 1$, 200 fixed surface/parameter settings per spillover regime. Application study: refined NC settings ($\rho = 0, 0.5$; $\gamma = 0.8$) in each year 2018–2021. Thin lines span the 5th to 95th percentiles of allocations, thick bars the 25th to 75th, and circles the mean; a design with one allocation per setting appears as a circle only.

The tail estimates remain uncertain. With $J = 100$, the worst decile contains only ten sampled draws, and extra outcomes reduce noise in $\hat{m}(a)$ but do not enlarge the allocation sample. All 112 reported settings in the tail refinement for designs with more than one possible assignment flag uncertain worst-decile membership. Under both arms, the overlap between the worst deciles identified by the two outcome halves (the Jaccard index: allocations in both divided by allocations in either) was 0–0.25 for Incidence-Guided Saturation, and approximately 61–86% of the raw variance of estimated risk across allocations was attributable to outcome Monte Carlo noise. Saturation Quadrants' worst-decile membership was similarly unstable. Small both-arms tail differences therefore do not establish superior downside protection. Checkerboard and High Incidence Focus each have only one possible assignment for these inputs, so their 90th percentile and worst-decile mean equal their conditional mean MSE. That equality does not imply low error or absence of outcome uncertainty.

In the matched sensitivity analysis that adds a hypothetical incidence baseline and adjusts for $\mathbf{X}$, the MSE of High Incidence Focus rises to roughly 3.68–4.44 times its primary value under both arms and 1.42–2.52 times under control-only spillover. When treatment follows incidence, adjusting for incidence leaves less residual information about treatment. Comparing rook and

queen contiguity at the same extreme combinations of ρ and γ leaves the regime-specific conclusions broadly unchanged. Alternative regional summaries change the regional ordering in some years; where an alternative summary gives the same ordering and ties, it gives the same set of possible assignments and so the same results. We retain the prespecified mean-rank rule rather than select a summary retrospectively. Table S20 reports the matched sensitivity ratios and Table S6 the regional ordering.



Figure S22: Allocation-specific conditional MSE in the refined primary queen setting $\rho = 0.5, \gamma = 0.8$, by year and spillover regime. Mean MSE, the estimated 90th percentile of allocation-specific MSE and the estimated worst-decile mean summarize the same $J = 100$ sampled assignments, with at least 400 independent outcomes per assignment of a randomized design or 1,000 for designs with a single possible assignment.

Table S20 reports Incidence-Guided Saturation's matched annual mean-MSE ratios. Rook means use only the four extreme combinations of ρ and γ shared with the primary analysis. Baseline-adjusted and regional-summary means use the six queen pairs. These are matched contrasts, not

ratios of unmatched pooled analyses.

Table S20: Incidence-Guided Saturation sensitivity MSE divided by matched primary MSE, by year and regime. Baseline adds rank-scaled $\mathbf{X}$ to both generation and fit; rook uses the matched extreme combinations of ρ and γ. Regional-summary alternatives preserve fixed regions.

Sensitivity	Regime	2018	2019	2020	2021
Baseline / adjusted	Both	1.016	1.030	1.015	1.035
Baseline / adjusted	Control	1.026	1.023	1.042	1.040
Rook contiguity	Both	1.005	0.998	0.985	0.998
Rook contiguity	Control	0.967	0.969	0.971	1.036
Mean regional rate	Both	1.034	1.000	1.000	1.031
Mean regional rate	Control	0.907	1.000	1.000	0.928
Pooled regional rate	Both	1.015	0.979	0.981	1.003
Pooled regional rate	Control	0.962	1.133	1.061	1.030

15 Application: participation, power and sensitivity

This section reports the participation, power and sensitivity analyses of the application. All numbers come from aggregate summaries of runs that wrote to their own output folders, each with a manifest of parameters, code hashes and package versions. These runs did not change the main application results. Unless stated otherwise, results use random volunteering, $\beta = 0$, $\kappa = 0.8$ and the corrected t test, and they average the four planning years.

15.1 Colleges needed and power

Under both-arms spillover, the design hardly affected the number of colleges a trial needs. Every design except Checkerboard and Isolation Buffer needed 35 to 37 colleges for 80% power to detect a college-level effect of one standard deviation, and about 19 colleges for 1.5 standard deviations (Figure S23, Table S21). Checkerboard and Isolation Buffer did not reach 80% power at $\tau = 1$ even with all 58 colleges.

Under control-only spillover, no design reached 80% power at $\tau = 1$ with all 58 colleges. At $\tau = 1.5$ the two saturation designs needed 28 colleges, against 29 to 32 for SRS and the rank-based designs, while 2×2 Blocking and Isolation Buffer failed to reach 80% power with 58 colleges in one of the four planning years. Power under control-only spillover rose slowly beyond about 40 colleges (Figure S24). In Figure S24 the two saturation designs nearly coincide, and the year-to-year range is narrower than the lines except near 58 colleges. Because each number of colleges uses its own participant subsets, the small drop in power from 29 to 30 colleges reflects which subsets were drawn. When nearly every college participates, nearly every control has a treated neighbor, so spillover exposure varies little and is hard to separate from treatment. Non-participating neighbors restore some of that variation, and so do the saturation designs. With all 58 colleges, the smallest effect detectable with 80% power was 0.76 to 0.78 college-level standard deviations under both-arms spillover for every design except Checkerboard (1.32) and Isolation Buffer (1.50), and 1.24 to 1.25 for the two saturation designs under control-only spillover, against 1.45 to 1.57 for SRS and the rank-based designs.

For a trial with $m = 25$ trained professionals per college and an ICC of 0.05, so that $\sigma_c = 0.30$, an individual-level effect of $d = 0.3$ gives power 0.71 to 0.72 with 29 colleges and 0.96 with 58 under

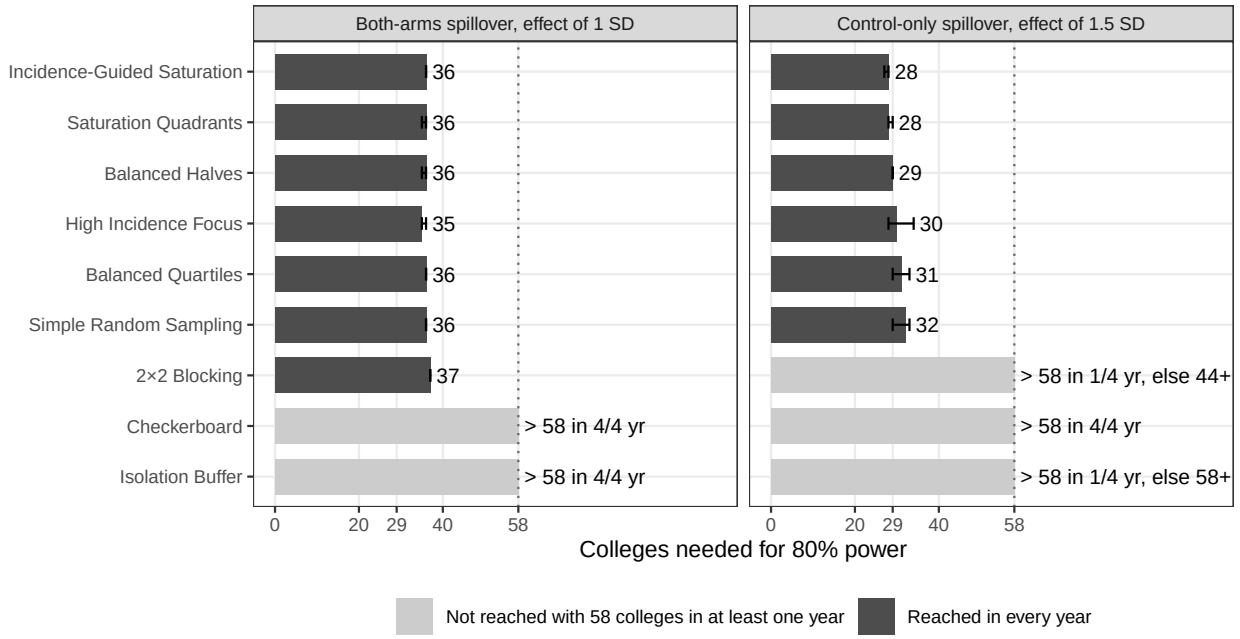


Figure S23: Number of participating colleges needed for 80% power with the corrected t test, by design, for a college-level effect of $\tau = 1$ standard deviation under both-arms spillover and $\tau = 1.5$ under control-only spillover, with random volunteering, $\rho = 0.5$ and $\gamma = 0.8\tau$. Dark bars show the mean over the 2018–2021 planning years and whiskers the range. Light bars mark designs that did not reach 80% power with all 58 colleges in the stated number of years, with the smallest number needed in the other years.

both-arms spillover, for SRS and for both saturation designs. Under control-only spillover the same effect gives power 0.49 and 0.53 with SRS, and 0.51 and 0.64 with Incidence-Guided Saturation (Table S22).

When the number of trained professionals varies, with m proportional to college population and averaging 25, and ICC = 0.05, power fell by 0.097 on average across designs, settings and effect sizes. For $d = 0.3$ under both-arms spillover, the mean power over the nine strategies fell from 0.59 to 0.46 with 20 colleges and from 0.88 to 0.77 with 58. MSE rose by a factor of 1.24 to 1.84 across designs and settings, and no design gained relative to the others.

The Wald test rejected a true null hypothesis in 11% to 13% of simulated trials with 15 colleges and in 6% to 7% with 58. With the corrected t test the rejection rate was 4.4% to 5.6%, and the coverage of the corrected interval at $\tau = 1$ was between 94.0% and 95.6% for every design and regime at every number of colleges (Table S23). The colleges-needed numbers above therefore come from a test with the nominal size.

Colleges needed and smallest detectable effect. Table S21 gives, for each strategy, the number of colleges needed for 80% power and the smallest college-level effect detectable with 80% power at 29 and 58 colleges. The number of colleges needed is interpolated monotonically between the simulated values of N within each year, averaged over years and rounded up to a whole college. Results at $\rho = 0$ differed from those at $\rho = 0.5$ by about two colleges or fewer. Power curves are not perfectly monotone in N within Monte Carlo error, because the participant subsets differ between values of N, and we report the dips rather than smooth them.

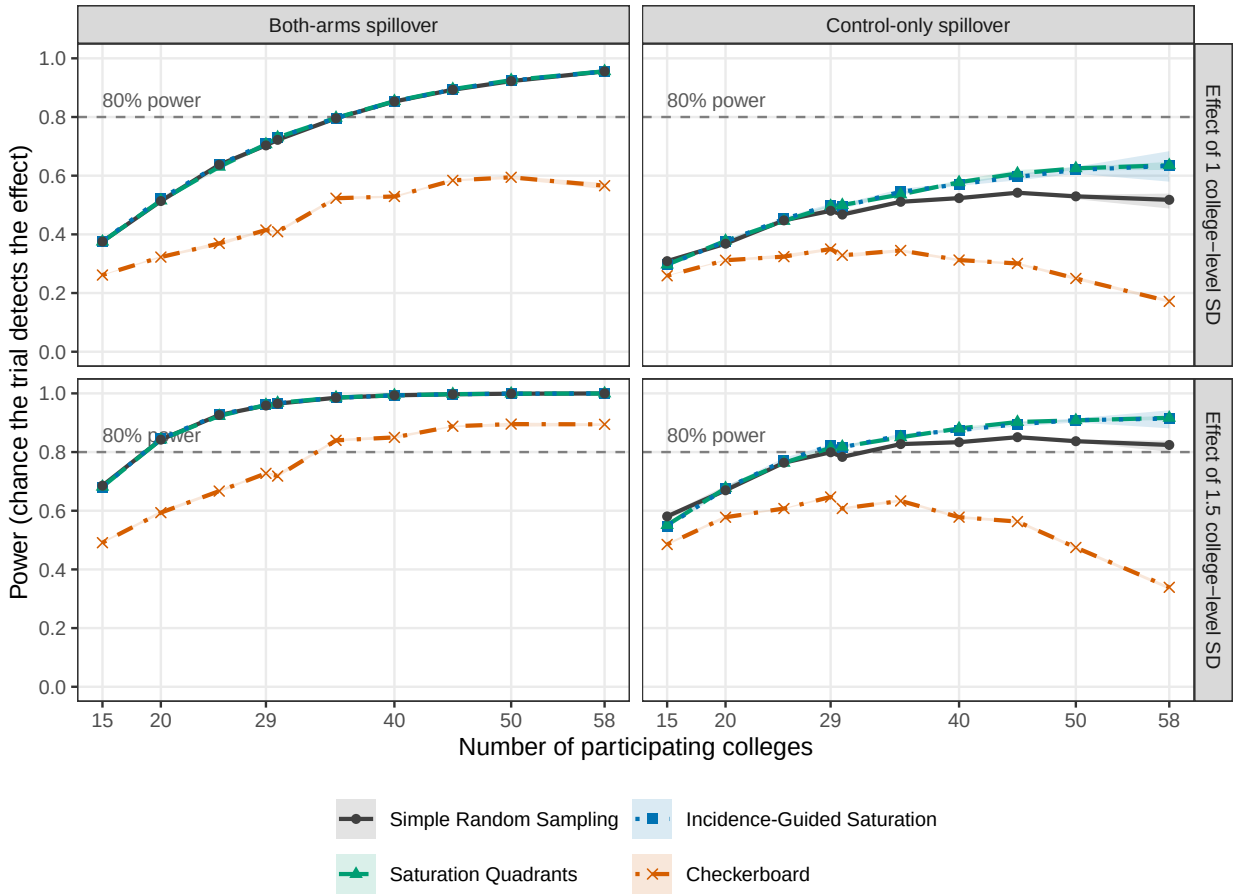


Figure S24: Power of the corrected t test against the number of participating colleges for simple random sampling (SRS), the two saturation designs and Checkerboard, by spillover regime (columns) and college-level effect (rows), with random volunteering, $\rho = 0.5$ and $\gamma = 0.8\tau$. Lines show the mean over the 2018–2021 planning years and bands the range, and the dashed line marks 80% power. Each number of colleges uses its own 50 random participant subsets.

15.2 Interval calibration

We compared several interval corrections and adopted a method only if its coverage lay between 94.0% and 96.0% for every design and regime. On the stored fits of the main application ($\beta = 0$), the Wald interval covered 92.6% to 93.6% by design and regime. A t reference distribution alone gave 93.2% to 94.1%, a degrees-of-freedom-corrected standard error alone 93.5% to 94.4%, and the two together 94.1% to 94.9%. In a separate run of 500 fits per setting at 58 colleges, parametric percentile bootstrap intervals with 499 replicates covered 90.1% to 93.6%, parametric bootstrap-t intervals 93.2% to 95.8% and profile-likelihood intervals 91.8% to 94.7%. Only the t interval with the corrected standard error met the rule on the stored fits. In the separate 500-fit run it covered 92.8% to 95.8%, narrowly missing the rule there, as every other method did. Table S23 shows its size and coverage at each number of participating colleges. Table S24 compares the Wald and corrected intervals for each design in the grid study and the main application, computed from the same fits.

Table S21: Colleges needed for 80% power and smallest effect detectable with 80% power (MDE, college-level standard deviations), corrected t test, random volunteering, $\rho = 0.5$, spillover $\gamma = 0.8\tau$, mean over the 2018–2021 planning years. > 58: 80% power not reached with all 58 colleges in at least one year. > 2: 80% power not reached at the largest simulated effect, $\tau = 2$.

	Colleges needed		MDE, 29 colleges		MDE, 58 colleges	
Design	Both arms $\tau = 1$	Control only $\tau = 1.5$	Both	Control	Both	Control
Checkerboard	> 58	> 58	1.70	1.86	1.32	> 2
High Incidence Focus	35	30	1.13	1.51	0.76	1.57
Saturation Quadrants	36	28	1.14	1.47	0.76	1.24
Isolation Buffer	> 58	> 58	> 2	> 2	1.50	1.50
2×2 Blocking	37	> 58	1.15	1.63	0.78	1.60
Balanced Quartiles	36	31	1.14	1.50	0.77	1.46
Balanced Halves	36	29	1.13	1.49	0.77	1.46
Incidence-Guided Saturation	36	28	1.13	1.46	0.76	1.25
Simple Random Sampling	36	32	1.14	1.50	0.77	1.45

Table S22: Power of the corrected t test for an individual-level standardized effect d with simple random sampling, by the number of trained professionals per college (m) and the intraclass correlation coefficient (ICC), with $\sigma_c = \sqrt{\text{ICC} + (1 - \text{ICC})/m}$ and $\tau = d/\sigma_c$. Random volunteering, $\rho = 0.5$, spillover $\gamma = 0.8\tau$, mean over the four planning years.

			Both arms				Control only			
			29 colleges		58 colleges		29 colleges		58 colleges	
m	ICC	σ_c	$d = 0.2$	$d = 0.3$	$d = 0.2$	$d = 0.3$	$d = 0.2$	$d = 0.3$	$d = 0.2$	$d = 0.3$
10	0.01	0.330	0.33	0.62	0.61	0.90	0.22	0.41	0.24	0.45
10	0.05	0.381	0.27	0.51	0.50	0.82	0.18	0.33	0.20	0.36
10	0.10	0.436	0.22	0.41	0.40	0.71	0.15	0.27	0.17	0.30
25	0.01	0.223	0.61	0.91	0.90	1.00	0.41	0.71	0.44	0.74
25	0.05	0.297	0.40	0.71	0.69	0.96	0.26	0.49	0.29	0.53
25	0.10	0.369	0.28	0.53	0.52	0.85	0.19	0.35	0.21	0.38
50	0.01	0.173	0.81	0.98	0.98	1.00	0.59	0.87	0.63	0.89
50	0.05	0.263	0.48	0.80	0.79	0.98	0.31	0.58	0.34	0.62
50	0.10	0.344	0.31	0.59	0.58	0.88	0.21	0.39	0.23	0.42

15.3 Guidance under partial participation

Table S25 gives each strategy's mean MSE relative to SRS at $\tau = 1$ for 20, 29, 40 and 58 participating colleges. The regime-specific guidance held with fewer participants, but the size of the saturation advantage depended on participation. Checkerboard and Isolation Buffer remained worse than SRS under both-arms spillover at every number of colleges, and the rank-based designs remained within a few percent of SRS. Under control-only spillover, the saturation designs' mean MSE was 22% below that of SRS with 58 colleges, 9% to 10% below with 40, 4% to 5% below with 29 and about 1% below with 20. With few participants, non-participating neighbors already create the exposure variation that saturation otherwise supplies. Isolation Buffer went from matching SRS under control-only spillover with 58 colleges to 2.3 times SRS's MSE with 29 colleges and 6.2 times with 15.

When colleges with higher incidence were more likely to volunteer, the ratios at 20 and 29

Table S23: Size and coverage of the Wald and corrected t tests by the number of participating colleges, random volunteering, $\beta = 0$, all designs, regimes, values of ρ and κ, and years. Size is the rejection rate of a true null hypothesis $\tau = 0$ (range over designs). Coverage is at $\tau = 1$, with the range over design-by-regime means in parentheses. All entries are percentages.

Colleges	Wald size	Corrected t size	Wald coverage	Corrected t coverage
15	11.1–12.8	4.4–5.4	87.7	94.9 (94.5–95.6)
20	9.5–10.4	4.7–5.4	89.8	94.8 (94.4–95.3)
25	8.4–9.5	4.7–5.5	90.9	94.8 (94.4–95.2)
29	8.3–8.8	5.0–5.5	91.6	94.8 (94.4–95.1)
30	7.9–8.6	5.0–5.5	91.7	94.7 (94.4–95.0)
35	7.6–8.1	5.1–5.5	92.2	94.7 (94.2–95.1)
40	7.4–7.7	5.2–5.5	92.5	94.7 (94.2–94.9)
45	6.9–7.5	5.0–5.6	92.7	94.7 (94.3–95.2)
50	7.0–7.3	5.2–5.6	92.9	94.6 (94.2–95.1)
58	6.3–7.0	5.0–5.5	93.2	94.6 (94.0–95.1)

Table S24: Coverage of the 95% Wald interval and the corrected t interval (t reference distribution with a degrees-of-freedom-corrected standard error) for the six designs and SRS. Grid: oracle model at $\tau = 1$, pooled over the five incidence configurations and both spillover regimes (160 scenarios per entry). Application: queen weights, $\beta = 0$, averaged over the four planning years and the six ρ/γ pairs of each regime. [a]Under rook contiguity on the grid, τ is not identified for Checkerboard.

	Grid, queen		Grid, rook		NC, both arms		NC, control only	
Design	Wald	t	Wald	t	Wald	t	Wald	t
Incidence-Guided Saturation	0.937	0.947	0.942	0.952	0.935	0.949	0.927	0.942
Balanced Quartiles	0.936	0.946	0.940	0.948	0.933	0.948	0.929	0.944
Isolation Buffer	0.939	0.949	0.939	0.950	0.932	0.947	0.931	0.947
High Incidence Focus	0.938	0.948	0.939	0.948	0.934	0.949	0.931	0.946
2×2 Blocking	0.939	0.948	0.938	0.948	0.932	0.947	0.929	0.944
Checkerboard[a]	0.941	0.950	0.150	0.162	0.936	0.949	0.933	0.948
Simple Random Sampling	0.938	0.948	0.938	0.947	0.932	0.947	0.929	0.945

colleges were within 0.02 of 1 for both saturation designs and Balanced Quartiles under either regime, and within 0.03 for Balanced Halves, so the saturation designs' advantage over SRS was 2% or less. Checkerboard and Isolation Buffer remained the worst designs. Rebuilding the four saturation regions on the participants left at least two colleges in every region by construction. Before that requirement was imposed, a region with fewer than two colleges occurred in 1 of the 50 random subsets at 15 colleges and in one incidence-weighted subset at 20 colleges. Imposing it changed the regions of only two distinct participant subsets and changed no ratio in Table S25 by more than 0.01.

15.4 Sensitivity analyses

The design comparison did not depend on the spatial model used to generate or fit the data, except for a Bayesian CAR model under strong spatial dependence. It also held under two other forms of

Table S25: Mean MSE relative to simple random sampling (SRS) at $\tau = 1$ by the number of participating colleges, random volunteering, mean over 16 blocks (year $\times$ ρ $\times$ κ). Below 1 is lower error than SRS.

	Both arms				Control only			
Design	20	29	40	58	20	29	40	58
Checkerboard	1.88	2.18	2.23	2.82	1.18	1.38	1.80	3.10
High Incidence Focus	0.98	0.97	0.96	0.92	0.99	1.00	0.99	1.06
Saturation Quadrants	1.00	0.98	1.00	1.00	0.99	0.96	0.90	0.78
Isolation Buffer	5.92	4.21	3.74	3.87	3.61	2.25	1.51	0.99
2×2 Blocking	1.03	1.01	1.02	1.03	1.08	1.11	1.11	1.18
Balanced Quartiles	1.00	0.99	0.99	1.01	1.00	0.99	0.99	1.02
Balanced Halves	0.99	0.98	0.98	1.00	0.99	0.97	1.01	1.01
Incidence-Guided Saturation	0.99	0.97	1.00	1.00	0.99	0.95	0.91	0.78
Simple Random Sampling	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

spillover exposure, and formal rank tests found a consistent ordering of the strategies within each regime.

Spatial model. Fitting a CAR model, generating the data from a CAR model, or both, left the saturation designs with 0.76 to 0.82 times SRS's MSE under control-only spillover, and Checkerboard and Isolation Buffer the worst designs under both-arms spillover. With the longer chains, 83.1% of the Bayesian CAR chains at $\rho = 0.5$ converged, against 46.2% for the shorter chains, but 12.5% still had an absolute Geweke z of at least 1.96, against about 5% expected by chance. Among converged chains, the Bayesian CAR estimate was biased by +0.14 under both-arms spillover and +0.22 under control-only spillover, with credible-interval coverage of 90.4% and 87.7%. For maximum likelihood on the same data, the biases were +0.02 and +0.03 and the corrected t coverage 95.0% and 94.6%. The Bayesian CAR model also did not reproduce the saturation designs' control-only advantage. It assumes a different dependence structure from the one that generated the data, and the correctly specified Bayesian spatial lag model does not share the problem. Of the Bayesian spatial lag chains, 94.9% converged. The median effective sample size of τ was 8,000 (minimum 5,897) and 5.1% had an absolute Geweke z of at least 1.96. Its posterior means correlated 0.999 with the maximum likelihood estimates (mean absolute difference 0.017). Its bias was at most 0.04 in absolute value in every combination of ρ and regime, and its credible intervals covered 94.6% to 95.3% over all chains (94.5% to 95.3% over converged chains). The Spearman correlation of its design ordering with that of maximum likelihood averaged 0.96 to 1.00 across the four combinations of ρ and regime (0.83 in the lowest single block), and the best design agreed in every block. Tables S26 and S27 give each strategy's MSE relative to SRS under each model.

Spillover form. Table S28 compares the three forms. Under the binary form, Isolation Buffer is never identified, because every control has a treated neighbor, and under control-only spillover no strategy was identified in every fit. The saturation designs were identified most often there (in 89.7% and 91.7% of fits) and, among identified fits, had MSE 0.70 to 0.71 times that of SRS. Under the count form, the saturation designs still had the lowest MSE under control-only spillover.

Formal tests on the NC results. Friedman tests over the 24 combinations of year, ρ and γ rejected equal ranks of the nine strategies in all four combinations of regime and β ($\chi^2 = 135.3$, 153.8, 150.7 and 179.8 on 8 degrees of freedom, $p \leq 2.3 \times 10^{-25}$; Nemenyi critical difference 2.45). Under control-only spillover, Saturation Quadrants and Incidence-Guided Saturation had the two

Table S26: Mean MSE relative to simple random sampling (SRS) under four combinations of data-generating model and fitted model: spatial lag (SAR) or conditional autoregressive (CAR). 58 colleges, $\beta = 0$, mean over 24 blocks (year $\times$ ρ $\times$ γ). Below 1 is lower error than SRS.

	Both arms				Control only			
Data	SAR	SAR	CAR	CAR	SAR	SAR	CAR	CAR
Fit	SAR	CAR	SAR	CAR	SAR	CAR	SAR	CAR
Checkerboard	2.88	2.83	2.93	3.09	2.99	2.64	3.01	3.00
High Incidence Focus	1.06	1.02	1.07	1.07	1.08	1.03	1.10	1.09
Saturation Quadrants	1.04	1.02	1.04	1.03	0.79	0.81	0.81	0.78
Isolation Buffer	4.11	4.13	4.16	4.45	1.03	0.95	1.03	1.05
2×2 Blocking	0.98	0.99	1.00	1.02	1.21	1.17	1.22	1.22
Balanced Quartiles	1.04	1.05	1.04	1.06	1.03	1.00	1.03	1.02
Balanced Halves	1.04	1.03	1.05	1.04	1.02	0.98	1.03	0.99
Incidence-Guided Saturation	1.03	1.04	1.04	1.03	0.81	0.82	0.81	0.76
Simple Random Sampling	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table S27: Mean MSE relative to simple random sampling (SRS) at $\rho = 0.5$ for maximum likelihood (ML), the Bayesian spatial lag model (Bayes SAR) and the Bayesian CAR model with converged chains only (Bayes CAR), all fitted to the same spatial lag data. 58 colleges, $\beta = 0$, $\gamma = 0.8$, mean over the four planning years.

	Both arms			Control only		
Design	ML	Bayes SAR	Bayes CAR	ML	Bayes SAR	Bayes CAR
Checkerboard	3.14	3.18	2.90	3.16	3.16	2.31
High Incidence Focus	0.96	0.96	0.83	1.19	1.19	1.04
Saturation Quadrants	1.12	1.10	1.11	0.86	0.86	1.05
Isolation Buffer	3.89	3.94	3.80	1.01	1.02	0.87
2×2 Blocking	1.14	1.14	1.11	1.48	1.48	1.34
Balanced Quartiles	1.00	1.00	1.05	1.10	1.10	1.15
Balanced Halves	0.97	0.97	0.99	1.05	1.06	1.01
Incidence-Guided Saturation	1.12	1.10	1.13	0.81	0.81	0.95
Simple Random Sampling	1.00	1.00	1.00	1.00	1.00	1.00

best average ranks (1.67 and 1.79 with $\beta = 0$; 1.29 and 1.71 with $\beta = 1$), and SRS ranked seventh with $\beta = 0$ and sixth with $\beta = 1$. Under both-arms spillover with $\beta = 0$, High Incidence Focus ranked first (1.88) and SRS sixth (5.17), with the balancing and saturation designs between them. These tests treat the 24 blocks as independent, although the years share one map and some strategies share their simulations across years, so they describe the consistency of the ordering rather than independent evidence.

15.5 Implications for trial planning

The application supports considering either saturation strategy when control-only spillover is plausible, and SRS alongside other randomized strategies when both-arms exposure is expected. A trial team should consider likely spillover, recruitment, cluster and participant budgets, achievable measurement precision and interval calibration before selecting one realized allocation. The findings do not establish intervention effectiveness or a final trial protocol.

Table S28: Mean MSE relative to simple random sampling (SRS) under three spillover forms, 58 colleges, $\beta = 0$, mean over 16 blocks (year $\times$ ρ $\times$ γ). n.i.: not identified in every fit (the spillover covariate is aliased with the intercept and treatment in some or all allocations).

	Both arms			Control only		
Design	Proportional	Count	Binary	Proportional	Count	Binary
Checkerboard	2.97	1.78	1.07	3.09	1.41	n.i.
High Incidence Focus	0.91	0.92	0.96	1.03	1.04	n.i.
Saturation Quadrants	1.01	1.02	1.03	0.76	0.81	n.i.
Isolation Buffer	3.82	3.84	n.i.	0.95	1.16	n.i.
2×2 Blocking	1.04	1.03	1.00	1.17	1.07	n.i.
Balanced Quartiles	1.01	1.01	1.00	0.99	0.99	n.i.
Balanced Halves	1.02	1.02	1.02	0.98	0.99	n.i.
Incidence-Guided Saturation	1.01	1.03	1.04	0.75	0.79	n.i.
Simple Random Sampling	1.00	1.00	1.00	1.00	1.00	n.i.

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